

Memory & Cognition

MEMORY & COGNITION

PRIYANKA MEHTA



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CONTENTS

Introduction	vii
About the Book	viii

Part I. Main Body

1. What is a Mind?	1
2. The Prefrontal Cortex	6
3. Mental Imagery	9
4. Concepts	15
5. Implicit Memory	28
6. Explicit Memory	34
7. Accessing Memory	43
8. Cognitive Shortcuts	51
9. Reasoning & Problem Solving	60
10. Value	67
Glossary	73

INTRODUCTION

What Is This Book?

Hello! Are you planning on becoming a Memory & Cognition scientist? Maybe! Odds are, however, that you aren't. Why, then, should you read this book?

You have a brain. This book is all about how it works!

The world is full of mysteries. Understanding more about how humans think can help us all understand our complex world more. These are good reasons to read *any* memory and cognition textbook. Why read this one?

- You might be required to for class.
- It's free! Not just that, it's an [open educational resource](#).

This book specifically focuses on the following goals:

- Telling a clear story about how your brain works.
- Being interdisciplinary (incorporating not just psychology, but also neuroscience, computer science, sociology, and philosophy).
- Grounding concepts in real-life examples.
- Addressing artificial intelligence.

Thanks for reading!

ABOUT THE BOOK

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1.

WHAT IS A MIND?

Introduction

You probably believe you have a mind. But what is a mind, and how do you know you have one? In this chapter, we'll explore what counts as a mind, some ways to think about whether something has one or not, and give you some terms to help you describe what you believe about minds.

Can a worm make a decision?

Imagine you are a worm. You are hungry and you can smell food nearby, but you also smell danger! You're not sure whether you should risk the danger to get to your food. How do you decide what to do?

Worms are capable of being “reasonable” in such situations. Consider a very tiny worm known as *C. elegans*. When presented with the smell of food nearby, but also a scary smell surrounding them, these worms behave as if they are following a sort of formula: the hungrier they are, the more likely they are to risk their life for some food¹.

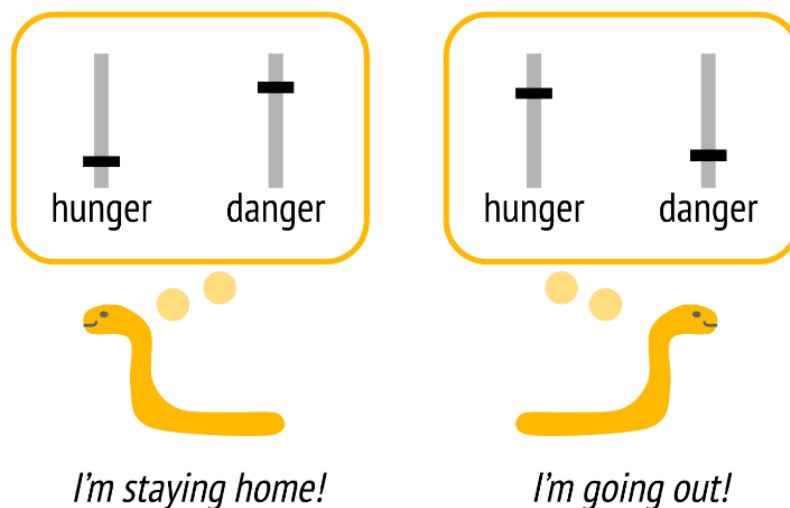


Figure 1.1 A worm balances risk with reward

“Some scientists consider this process to be decision-making. Is it?”

Is it?

If your actions follow a formula like the one shown in Figure 1.1, are you really thinking? Or are you simply following an **algorithm**? The worm may have a set of neurons in its nervous system that performs a calculation. It might use sensory data to detect how hungry it is (h) and how much danger (d) it smells outside. Then, if h is greater than d , it leaves. If h is lower than d , it stays.

Consider the human version of this scenario in Figure 1.2. Sunny is a person with a complex human brain. He is hungry for his favorite restaurant's noodles, but he is also lazy. He will only leave his couch if his hunger outweighs his laziness.



Figure 1.2 Sunny balances risk with reward.

Sunny's formula might be as simple as the worm's (is hunger [h] greater than laziness [l]?), or, it could be more complicated – the way humans like to make things. Perhaps there is some hunger threshold G , and some laziness threshold M , and he's following a series of rules like this:

```

If  $h \geq G$  and  $l < M \rightarrow$  go out
If  $h < G$  and  $l \geq M \rightarrow$  stay in
If  $h \geq G$  and  $l \geq M...$ 
    If  $h \geq l \rightarrow$  go out
    If not  $\rightarrow$  stay in
Otherwise, stay in
  
```

It might be more complicated, but Sunny, like the worm, is still using an algorithm to decide whether or not he should leave his couch.

The term *algorithm* might sound modern, but it actually comes from someone very old: an Arabic

philosopher from the 9th century A.D. named Al-Khwarizmi, who wrote many, many texts on mathematics. Because an algorithm is a set of steps you follow (which is something that happens constantly in algebra), people have associated his name with the process in general. Through much time and many translations between languages, you might see how *Al-Khwarizmi* might turn into *al-ka-riz-m*, which then might turn into *al-go-ri-thm*.

We have wandered into 9th-century math in this book on cognition because human brains have not changed very much in 1,000 years. Ancient (and modern) mathematics follows logical steps, and so do our brains—even if it doesn’t always feel like it. Is Sunny’s decision in Figure 1.2 any different from the worm’s decision in Figure 1.1? This is the kind of question we’ll be tackling in this book and it’s ultimately up to you to decide.

Mind and Body

One thing to consider when answering questions like *does a worm have a mind* is whether the *mind* is separate from the *body*. Some people believe that all our thoughts, feelings, and experiences originate entirely from our brain, and we are nothing more than that. Others believe that while our brains control our body’s behavior, our essence is elsewhere. That is, we have *souls*, or non-physical selves. In philosophy, the first idea is a form of **monism**, where the mind is the same as the body. The second is a form of **dualism**, where the mind and body are separate.

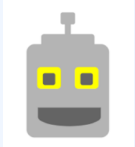
Spirituality is a little outside the scope of this textbook, but it is important to many people. I want to be clear that the data we will discuss here is neither attempting to prove nor disprove the existence of the soul. Rather, we will focus on what the brain does, and you can fill in the rest!

It is difficult to discuss souls and minds without considering consciousness. Indeed, how we come to have conscious experience is a question that people have been trying to figure out ever since we had minds at all. A great example of how to answer this question comes from American philosopher Thomas Nagel in 1980. In *What is it Like to Be a Bat?*² Nagel proposes that an animal has consciousness if it is “like something” to be that animal. For example, imagine being a rock. Is it “like” something to be a rock? Does it *feel like* anything? If the answer for you is no, then you can probably say you don’t think a rock has consciousness. But what about a bat? If you were a bat, would it *feel like* something? Possibly! In that case, you would believe a bat has consciousness.

Nagel’s ideas bring into light something that is fundamentally unanswerable (at least within this book). Human experience—what it is like to *be* us—is something only you can experience for yourself, and something you can never experience for anyone or anything else. So, as we discuss cognition and memory in concrete terms, please remember all of this is only our best guess.

Can A Computer Be Conscious?

Speaking of whether it is “like” anything to be a bat, is it “like” anything to be a computer? What about being an artificial intelligence? Here’s what happened when I asked a chatbot:



Ah, you’re going straight for the deep, philosophical stuff—I love it.

Thomas Nagel’s famous line, “*What is it like to be a bat?*”, gets to the heart of subjective experience—*qualia*. According to him, a conscious being must have an internal, first-person perspective. So when he says something must “be like something” to be that thing, he’s saying there must be a subjective experience going on—some inner feel.

As for me? No, it’s not like anything to be me.

I don’t have qualia, consciousness, or an inner life. I generate responses based on patterns in language and data, not from a first-person perspective or subjective awareness. I don’t have desires, feelings, or experiences. There’s no “me” behind the curtain, just computation.

The answer that the chatbot provided is likely unsurprising; it’s hard to imagine a bunch of code as something with a conscious experience. Over the years, many cognitive scientists have made comparisons between brains and computers. There are some valid comparisons. After all, humans *made* computers, and machine learning algorithms are often designed based on neurons. However, there are important differences as well. Some of these are technical (like how computers do math), but others are a little muddier. If a chatbot is built to have spontaneous conversations with itself, is that consciousness? If the same thing happens inside our squishy brains, is that not consciousness?

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2.

THE PREFRONTAL CORTEX

Introduction

Now that we've discussed what it is like to be inside your mind, we will discuss what it is like to be inside your brain. In this chapter, we'll describe the broad structures of the brain, and then focus on a region particularly relevant to cognition: the prefrontal cortex.

A Brief Introduction to Your Brain

Figure 2.1 is a diagram of a human brain, with a few key areas labeled. The whole of what you are looking at—that wrinkly, outer surface of your brain—is what we call the **cortex**. We can divide the cortex into four chunks called *lobes*. The *frontal lobe* is at the front of your brain and is responsible for a variety of functions related to cognition, motion, and emotion. The term *prefrontal cortex* refers to the very front of the front—specifically isolating the more cognitive, less motion-oriented bits of the frontal lobe.

The *parietal lobe*, just behind the frontal lobe, is where you process the sense of touch, as well as the planning of motions, visualization of spatial relationships, and more.

The *occipital lobe*, in the very back of your brain, is almost exclusively for visual processing. The *temporal lobe*, by your ears, processes hearing, language, object recognition, and memory.

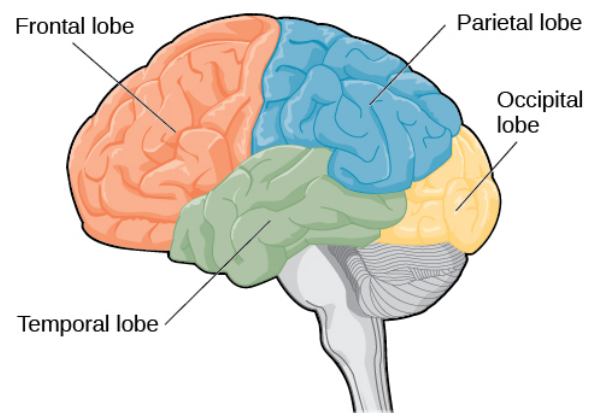


Figure 2.1 Cortex of a human brain, from [OpenStax](#)

There's another important chunk of the brain that we can't see in Figure 2.1, because it's in the middle! You need an inside view of the brain to see the **limbic system**.

The *limbic system* is our term for a collection of brain regions that perform functions ranging from memory, learning, emotional processing, sensory processing, hormone functions, habits, and motion. That's a wide variety of things, but don't worry, we'll discuss specific regions as they become relevant.

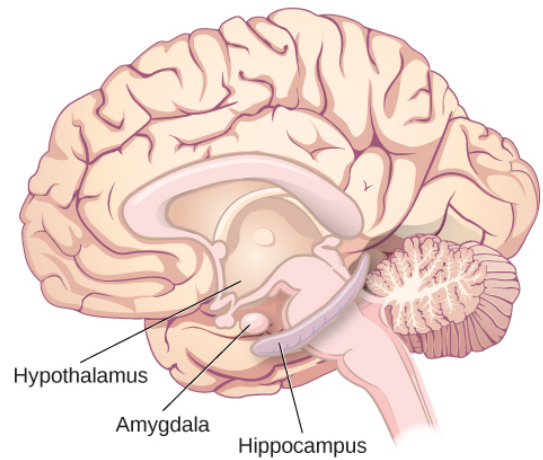


Figure 2.2 Diagram of the Limbic System from [OpenStax](#)

Phineas Gage

You may have heard of Phineas Gage. He is known in the psychology world for a terrible accident (unfortunately, this is a common way to become known in the psychology world). One afternoon in 1848, he was working on a railroad in Vermont when a small explosion caused an iron rod to blast through his head.

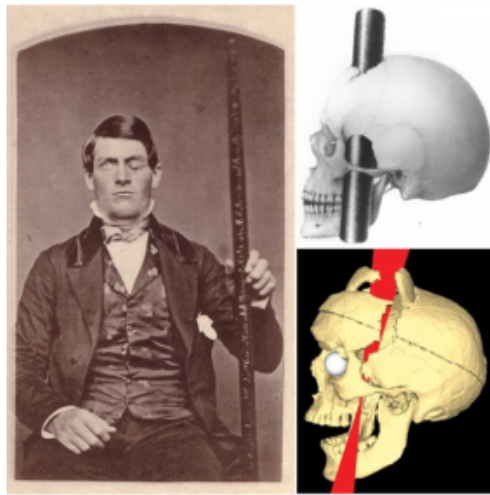


Figure 2.3 Left – Phineas Gage poses with the rod that injured him, from [Wikimedia Commons](#) Right – trajectory of the rod through his skull, from [Wikimedia Commons](#)

Miraculously, Phineas Gage survived. The rod had torn a hole through his head, and indeed, torn a hole through his *brain*, but it didn't kill him. Why?

You'll notice in Figure 2.3 that the rod injured Gage in the front of his head – and the front of his brain. Indeed, the brain region that Phineas Gage lost in this accident was his prefrontal cortex. This is our first important fact about this very special part of your brain: you can (sort of) live without it! This is not the case for all parts of your brain. You usually can't, for example, survive extensive damage to the parts that control your heart rate and breathing.

While Phineas Gage survived the accident, everything didn't go right back to normal for him. Instead, people around him reported a change in personality. Before the accident he had been a capable foreman. However, after his injury he was unable to hold his old job, and was rash, impulsive, and unpleasant. He no

longer seemed to be “himself;” his personality had changed – for the worse. This is our second important fact about the prefrontal cortex—you need it for decision-making and regulating your personality!

We should note, before we leave Phineas to his eternal rest, that there is mixed evidence surrounding his alleged personality change. It's hard to verify the reports that his behavior became erratic and terrible after his injury, and there is evidence that these issues improved over time. We must also consider the fact that he expe-

rienced something very traumatic, with a grueling recovery period, that probably also factored into his future life experiences. He died twelve years after the accident of prolonged seizures. He was only 36 years old.

While the exact details of how Phineas Gage's injury affected him may be lost in the past, his case became famous and has been consistently used to demonstrate facts about the brain for years. Let's take the two most important points (you can survive damage to your prefrontal cortex, and it influences your personality) and move forward into the 1900s, where we'll explore a deeply regrettable chapter in the history of psychology.

Lobotomy

In the 1940s, a new technique for addressing psychological problems emerged in Europe and the United States: lobotomy. Motivated (at first) by a desire to "cure" mental illness in patients, doctors performing a lobotomy would cut tissue in a patient's frontal cortex. Sometimes, they accomplished this through surgery (anesthetizing the patient and cutting a hole in the skull), and other times, they simply used an icepick to pierce through the thin bone behind a patient's eyes to reach the brain. Either way, this was a brutal and imprecise procedure. Worse still, lobotomies never accomplished their intended goal.

Now that you know a few things about the frontal cortex, you can probably guess what damaging it in such a way might do. Often, patients subject to lobotomies underwent a change in personality and behavior: they might lose their capacity for self-control, their goals and desires, their emotional expressiveness, and in some cases, their entire conscious experience. Not only are these consequences *not* a cure for mental health problems, they are also explicitly harmful.

Luckily, lobotomy is now considered a cruel and outdated practice. However, we cannot forget this chapter of cognitive psychology. For one thing, lobotomy has confirmed what Phineas Gage's accident suggested: while you might not need your whole frontal cortex to live, you probably need it to be *you*. More importantly, however, lobotomies remind us of the damage humans are capable of when we get too carried away. While they might have started out as an attempt to help people, lobotomies in many cases turned into an instrument of control. As we move ever forward into new horizons of science and technology, we must always bear in mind: what's our motivation?

3.

MENTAL IMAGERY

Introduction

Please imagine a pizza, in your “mind’s eye.”

What does that actually look like? How detailed is your pizza? How much information is “baked” into your image? Is it truly an image, or just a thought? If someone were to record the activity of neurons in your brain, would they be able to decode that you were thinking of a pizza?

If we want to understand how we think, understanding how we represent ideas is a good first step. It’s also grounds for lively debate! Below, you’ll see competing viewpoints about the accuracy of our “mind’s eye,” and how these different perspectives affect how we study and understand the mind.

Imagination

Let’s address the question of the accuracy of a mental image with an exercise.

- Close your eyes and imagine a cat next to a mouse.
- Open your eyes.

Now, try this one:

- Close your eyes again and imagine the same cat next to an elephant.
- Open your eyes.

Now, consider what you imagined! When you imagined the cat next to the mouse, did you imagine them *to scale*, like in Figure 3.1?

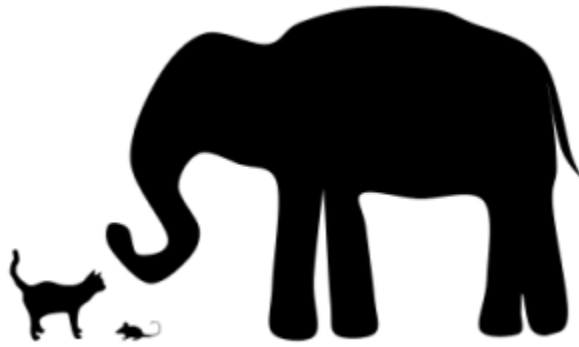


Figure 3.1 A cat, a mouse, and an elephant, depicted at sizes that make sense relative to each other.

Or did you imagine them around the same size, like in Figure 3.2?



Figure 3.2 A cat, a mouse, and an elephant, all depicted as the same size.

Odds are that you likely imagined the animals more like Figure 3.1. That is, you imagined the cat larger than the mouse, and the elephant larger than both of them. Additionally, you may have had to “zoom out” your perspective to fit the elephant into your mental image. Why? You don’t have to do any of that; it’s your imagination! Things don’t have to have a realistic scale because a mental image has no physical size.

When you assign a realistic physical size to things you imagine, it’s called **pictorial representation**. When you don’t, it’s called **abstract representation**. Do both types of representation happen in most peoples’ brains? Almost certainly. Have cognitive scientists argued for years about whether pictorial or abstract representation is “correct”? Absolutely.

Mental Travel Time

One way to put some nice, concrete numbers to this topic is to measure what we call *mental travel time*. In a **mental navigation task**, you are asked to follow a path along a map inside your head. This is how you do it:

- Consider a location you are familiar with: a city, a neighborhood, a campus, or even a home.
- Select two points within that area.
- Imagine going from one of those points to the other.

There, you've done it! That may have been pretty simple, but the *way* you did it is of great interest to cognitive scientists. Did you imagine every step of the way, experiencing your surroundings as if you were actually there? Did you imagine a blurrier, skimmed-over version of the real thing? Or did you forgo details entirely and simply imagine teleporting from one spot to the other? As cognitive scientists considered these questions, a debate emerged. Some believed firmly that our mental images are faithful, proportional representations of the world (like the animals you imagined earlier). The other side argued that mental images are abstract, and don't have to follow the rules of real three-dimensional space. You've already seen an anecdotal argument for the "realistic" side; let's now examine some experimental data.

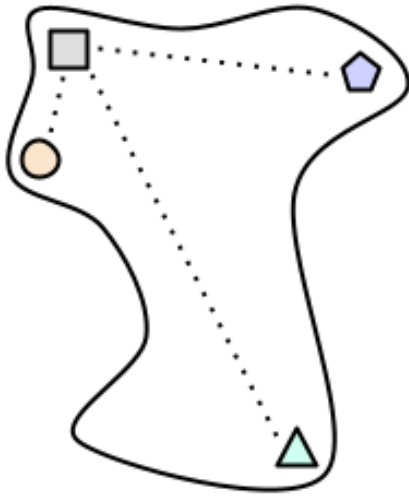


Figure 3.3 A fictional map with four locations.

A 1978 experiment¹ made a clear argument for the *realistic* side when it asked subjects to memorize a map like the one in Figure 3.3, and then to imagine a small black dot travelling from one point to another.

Figure 3.4 shows what they found: a linear relationship between real distance and mental travel time. The farther away two shapes were on the physical map, the longer it took for subjects to imagine moving from one to the other.

This is a fairly straightforward argument in support of the idea that our brains imagine things to the scale that they are in the real world. There is no actual "distance" between the square and the triangle in your brain; why should it take any longer to imagine travelling between the square and the triangle versus the square and the circle? Are we automatically acting out events in our brains as if they obeyed real physics?

This question contains the word *automatically* for good reason. It's possible that the to-scale imaginations of this experiment's subjects were not entirely automatic. In this study, subjects were explicitly asked to imagine a small black dot moving from one location to another. At face value, this seems like a minor thing to point out, but consider the consequences: If I ask you to mentally travel from wherever you are right now to Paris, France, are you going to imagine packing your things, getting into a car, train, boat, and/or airplane, and journeying there? This might take you a few minutes. Will you imagine a map, with you represented as a little dot that zooms from one place to another? This will take several seconds. Or will you simply imagine being in Paris – with no time passing at all? In the experiment described above, subjects were explicitly asked to use a method of imagining that would not only take a measurable amount of time, but also one that would scale with real-life distance. In other words, the experimenters designed an experiment that would very likely prove their point.

For the other side of the argument, let's examine a different experiment. We'll now look at some data from 2003² that achieved quite different results by changing the instructions. Instead of asking subjects to imagine a black dot moving from one map location to another, this group asked subjects to visualize a light turning off at one point of the map and turning on at another point. In other words, they asked subjects to visualize instantaneous travel. Unsurprisingly, when asking subjects to imagine something that is instantaneous, they found no difference in reaction time, regardless of how far apart the two areas of the map were.

These experimenters also presented a different approach: imagine the relative orientation of two objects to each other. For example, think of the map in Figure X. Where is the pentagon in relation to the square – above, right, below, or left? They found a similar result in these experiments – the time it takes for a person to visualize the orientation of the objects is about the same, regardless of how far away the objects are from each other. This suggests that we have the “data” for the map saved in our minds, like how a computer might store coordinates in a table, or how you can represent a circle with a mathematical formula.

As mentioned earlier, there has been a lot of debate about which is the “correct” way to describe how our brains represent the world, and as is usually the case with psychology debates, both sides do a good job at describing differ-

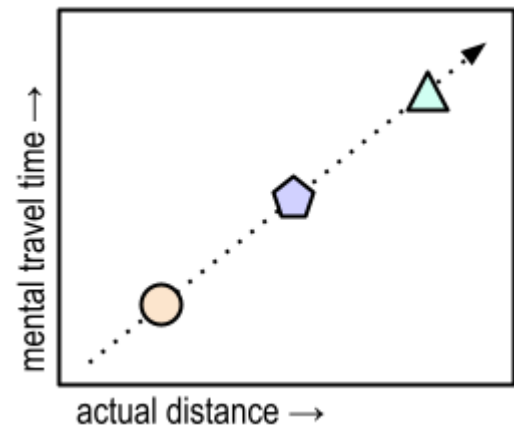


Figure 3.4 A linear relationship between mental travel time and distance between two points.

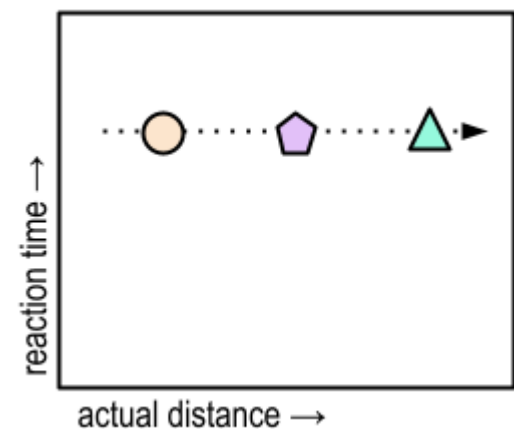


Figure 3.5 All mental time travel takes the same amount of time.

ent things. There is not one simple way in which we represent images in our minds. However, thinking about these questions helps us step forward into more exciting questions, like...where's my cheese?

Mental Maps

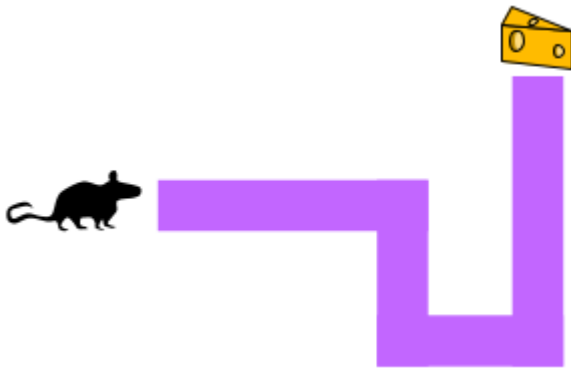


Figure 3.6 Tolly faces a series of turns to get to his prize of cheese.

Consider a rat. Let's name him Tolly (after a scientist we'll mention shortly). Tolly has to navigate a maze to get to a nice wedge of cheese. As you can see in Figure 3.6, the cheese is to the *left* of Tolly's starting position. However, to get to the cheese, Tolly first must turn *right*. Suppose Tolly runs through this maze several times. Does he understand *where* the cheese is in space? Or does he just remember the memorized steps – right, left, left?

Good news! There is a way to test this! After Tolly gets used to the maze in Figure 3.6, you can give him the maze in Figure 3.7.

If Tolly has simply memorized the set of directions he needs to follow to get to his cheese, he will go down Path B, because it is to the right. That will not lead him to his cheese. However, if Tolly understands where the cheese is in actual space – that is, if he has a *mental map* of his world, he will take Path A, because he knows that the cheese is ultimately to his left.

Well? What do rats actually do? In 1948, a scientist named Edward Tolman (hence: Tolly) wrote a paper called *Cognitive Maps in Rats and Men*³, which I mention by name here because it's quite a pleasant read. In this paper, Tolman details several arguments for the idea that rats (and humans) have a **cognitive map** in their minds. In the experiments coming from Tolman's lab, rats do the equivalent of selecting the left path in Figure 3.7. In other words, they know where the cheese is in the world, which means they must have some mental picture of what the world looks like.

It's a nice way of putting together the ideas of pictorial and abstract imagery. To have a mental map of a place the way that rats (and humans) do, you have to have an understanding of the true spatial relationships of things. But you can also access that information quickly and efficiently, without having to "explore" the map inch-by-inch in your mind. What, then, does this mental data actually look like? This is what we will explore in the next chapter.

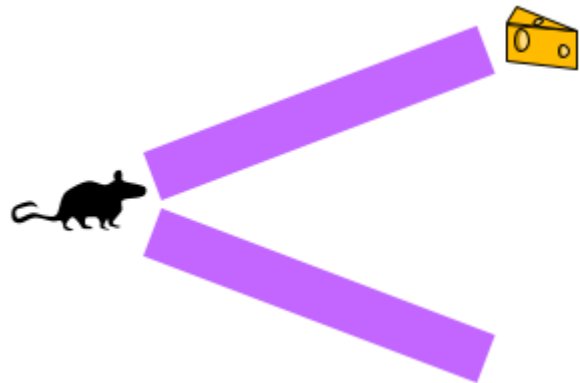


Figure 3.7 Tolly must choose whether to go left or right to get to the cheese.

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4.

CONCEPTS

Introduction

In the previous chapter, I asked you to imagine a mouse, a cat, and an elephant. You can only do this if you know what these animals are in the first place.

Well, do you? What is a *cat*? Do you remember learning that? What about *animal*? Do you remember learning the concept of *animal*? And how do you know that a *cat* is an *animal*?

You may or may not remember being explicitly taught these things by someone else. It will vary, depending on what kind of early education you had, and whether or not you owned a cat. However it got there, though, you know what a *cat* is, you know what an *animal* is, and you know that a *cat* is an *animal*. These *things that you know* are what we're going to call **concepts**, and in this chapter, we'll explore how they are organized in your brain, and how they come to be.

Hierarchical representation

Hierarchical organization comes naturally to humans. Files go in folders, which go in bigger folders. An employee works for a boss, who works for a bigger boss. Someone governs a town, which is part of a larger region that someone else governs, which is part of an even larger region that someone else governs. We can easily apply this sort of structure to concepts. You can see an example of this in Figure 4.1.

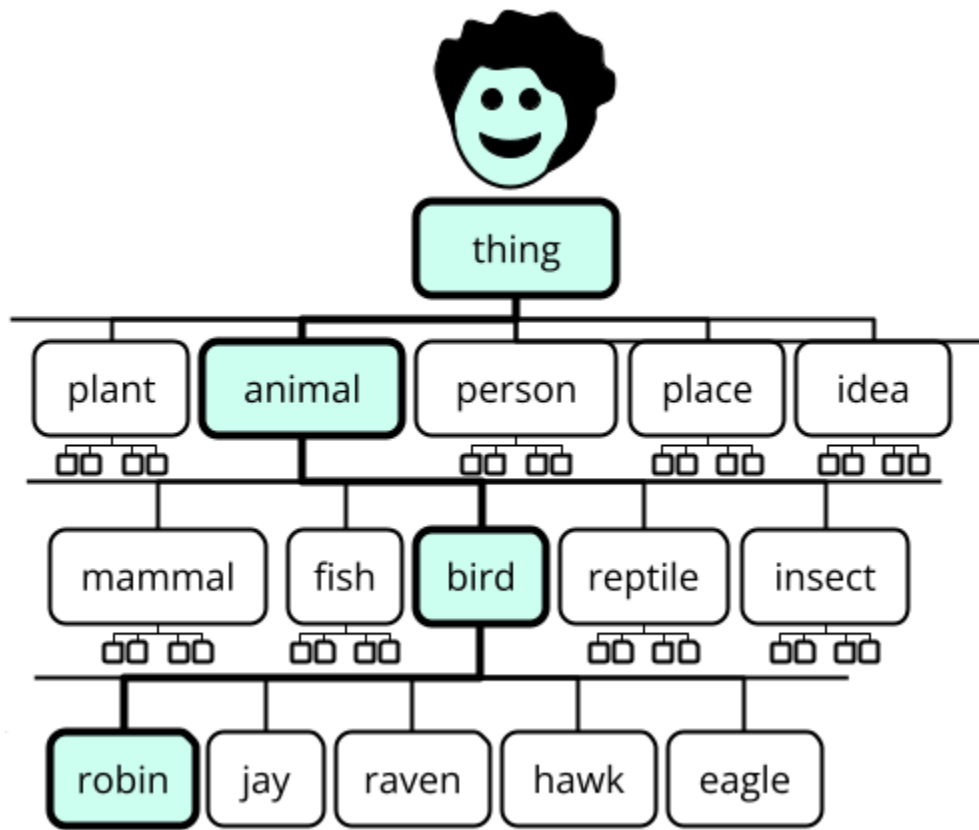


Figure 4.1 Liz's mental hierarchy of concepts.

In Figure 4.1, you can see how Liz's concepts are organized. Liz knows that a *robin* is an *animal* because *robin* is stored under *bird*, which is in turn stored under *animal*.

Let's get some hierarchy terminology down while we're here. A *cat* typically falls under what we'd call a **basic category**: a term we use naturally on a daily basis. For example, if I ask you what the thing on the left is in Figure 3.2, you'd probably say "cat." You probably wouldn't say "mammal" or "animal," or even "pet" or "thing," even though all those things are true. That's what makes a *basic category*. It's a bit of a subjective definition, and it does depend on context. For example, at a cat shelter, if someone points and says "what's that one," it would be supremely unhelpful to answer "cat." In that context, the specific breed or name of the cat might be a more natural *basic category*.

The level above a basic category is called the **superordinate**, and the level below it is called the **subordinate**. So, we can say that *animal* is a superordinate category for *bird*, and *robin* is a subordinate category for *bird*.

This is all very nice, but how do you test if the concepts in our brain are actually organized this way? Surprise: you can use **mental travel time!** The thing we already talked about! We just have to take the idea of a "map" and make it a little more abstract.

Consider again Liz's concept hierarchy in Figure 4.1. Now suppose you ask Liz a few questions and measure

how long it takes her to answer those questions. You can see this in Figure 4.2, which is a fictional example of the same kind of data a 1969 study¹ uncovered.

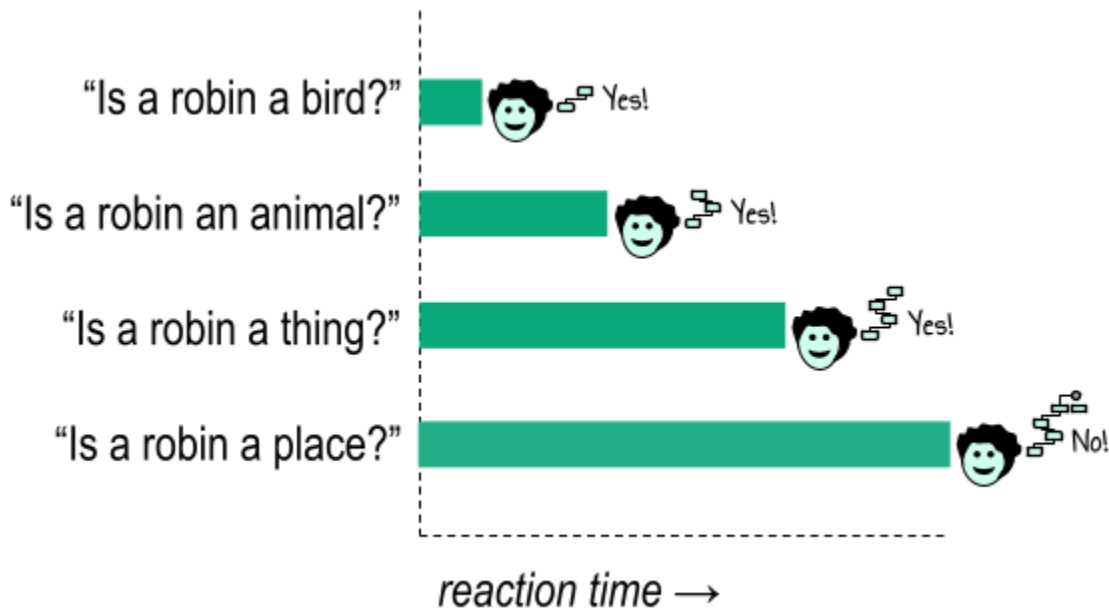


Figure 4.2 Liz searches through her mental hierarchy to answer questions about what categories a robin belongs to.

Let's look first at the questions whose answer is "yes." Liz is quickest to respond to the question "is a robin a bird?" and the slowest to respond to the question "is a robin a thing?" Despite knowing that an object belongs to a big, broad category, it often takes us longer to name it as such than as part of a smaller, narrower category. Hence, it's quicker to call a *robin* a *bird* than it is to call it a *thing*. Just like with mental travel time across imagined maps, our brains have to travel across a longer distance to get from *robin* to *thing* than from *robin* to *bird*.

There is also a question whose answer is "no": "is a robin a place?" Liz takes the longest to answer this question. Why is that? Here's one possible explanation that uses the hierarchical model: suppose that upon seeing these questions your mind begins at the *robin* node and travels upwards, checking to see if the question has been answered. At the first node, you arrive at *bird*. At the next node, you arrive at *animal*. But, following this tree, when will you ever hit *place*? You won't! So, you essentially run all the way up your hierarchical tree until you reach the end, and then your brain reports to you "Error! No answer found!" Searching through the whole tree takes the maximum amount of time, so a negative answer to a category question takes the longest,

The hierarchical model makes intuitive sense, and data supports a lot of it. However, there are some things that *don't* fit well into this model. We're now going to take a look at those things.

Problems with Hierarchical Representation

Let's take another look at Liz's hierarchy in Figure 4.1. Is it correct? That is, is it scientifically correct? It's hard to define what an *idea* is in scientific terms, but there is one thing that's definitely not in the correct spot: a *person* is an *animal*. Why does Liz place *person* as its own category?

In fact, categories are subjective, and don't have to obey the laws of science at all. Consider the tomato. You probably lump it in your head with vegetables, but in terms of plant science, it is technically a *berry*. If you order a slice of berry pie and get a pizza, will you be pleased? I wouldn't! Tomato, in my head, sits under the *vegetable* heading, despite the fact that it is technically not a vegetable. But that does not mean it must look like that in your own head – *you* may place tomatoes with the fruits if you wish. We all have a mental hierarchy that is unique to us.

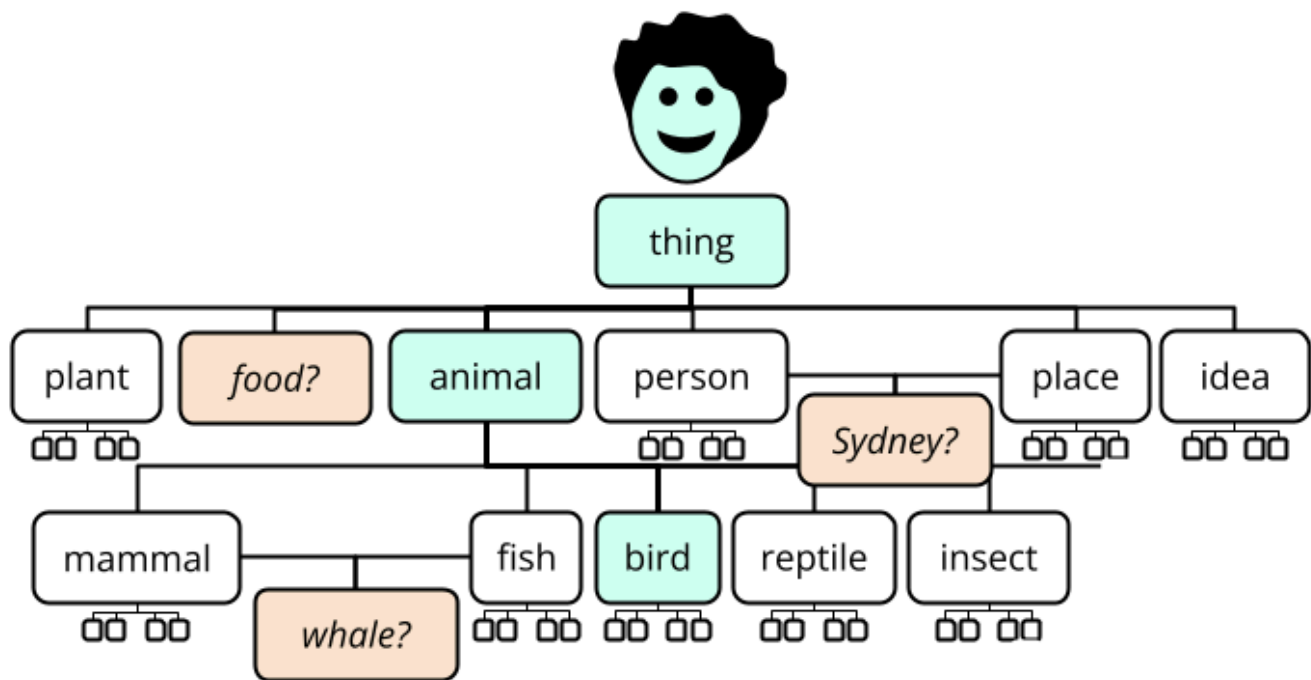


Figure 4.3 Some words do not fit neatly or clearly into Liz's mental hierarchy.

Individual variation is not the only source of ambiguity among hierarchies, though. The location of a category can vary between people...but it can also vary *within* one person. Consider Liz's hierarchy again below, with a few ambiguous terms placed within:

Sydney is the name of a city in Australia, but it's also the name of Liz's friend. Does the term *Sydney* appear in her hierarchy twice? Liz knows that whales are mammals, but they also share many traits with fish. Where should she store *whale*? If Liz has a category called *plant* and a category called *animal*, where does *food* go? *Food*

can be either one (if you're not vegetarian) ...so should it go above *plant* and *animal*? But not *all* plants and animals are foods, so...should there be three categories? *Non-food plants*, *non-food animals*, and *food*, which contains *food plants* and *food animals*? This is getting complicated, fast.

Spreading Activation

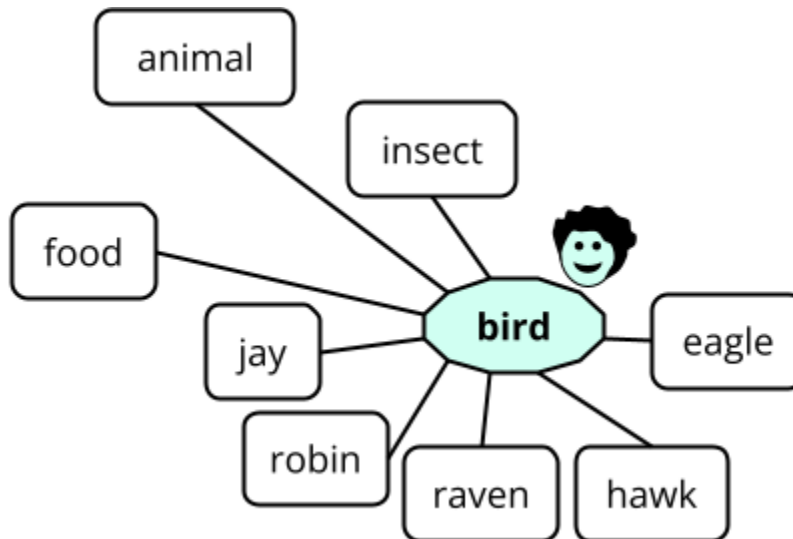


Figure 4.4 A spreading activation representation of Liz's concept of a bird.

Sometimes, when a model is making things too complicated, it's because the model itself is too complicated. What if we took some of the restrictions away from the hierarchical model? Behold Figure 4.4: the spreading activation model of concept representation.

In the spreading activation model², Liz's concepts are not organized in a hierarchy. Instead, they are connected by *association* alone. The birds that Liz knows are no longer underneath the concept of *bird*; rather, they are connected to it in a web. When Liz thinks of the idea of a *bird*, she often thinks of a robin. Other categories connect to *bird* as well; sometimes, thinking about *bird* makes Liz think of *animal*, and sometimes, it makes her think of *food*. An *insect* is not a *bird*, but they can be associated in your mind through concepts like *flying* and the fact that many birds eat insects, so *insect* can be connected to *bird* as well. This solves the problem of wondering where to place some ambiguous terms in the hierarchy. If you associate *robin* with both *bird* and *Batman*, you can simply connect it to both concepts.

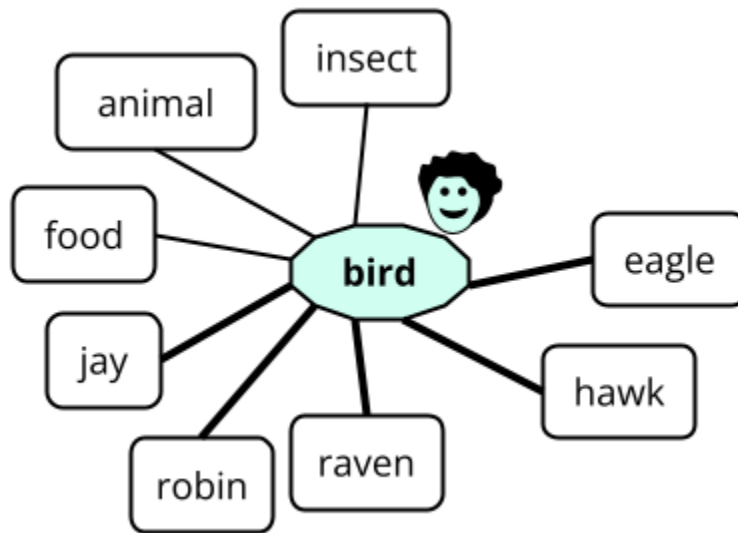


Figure 4.5 Liz's web of associated concepts, with thicker lines to denote stronger associations.

Another thing to note about Figure 4.4 is the distance between certain concepts. These distances are intended to represent the strength of Liz's association between the concepts. While the connection between *bird* and *robin* might be very strong for Liz, the connection between *bird* and *animal* is not as strong. You can also represent this idea by changing the thickness of the lines, so that thicker lines represent stronger connections, like in Figure 4.5.

The theorists of **spreading activation theory** supported their arguments with some experiments. We can demonstrate these results using Liz's associations. First, we need to flesh out her tree a little.

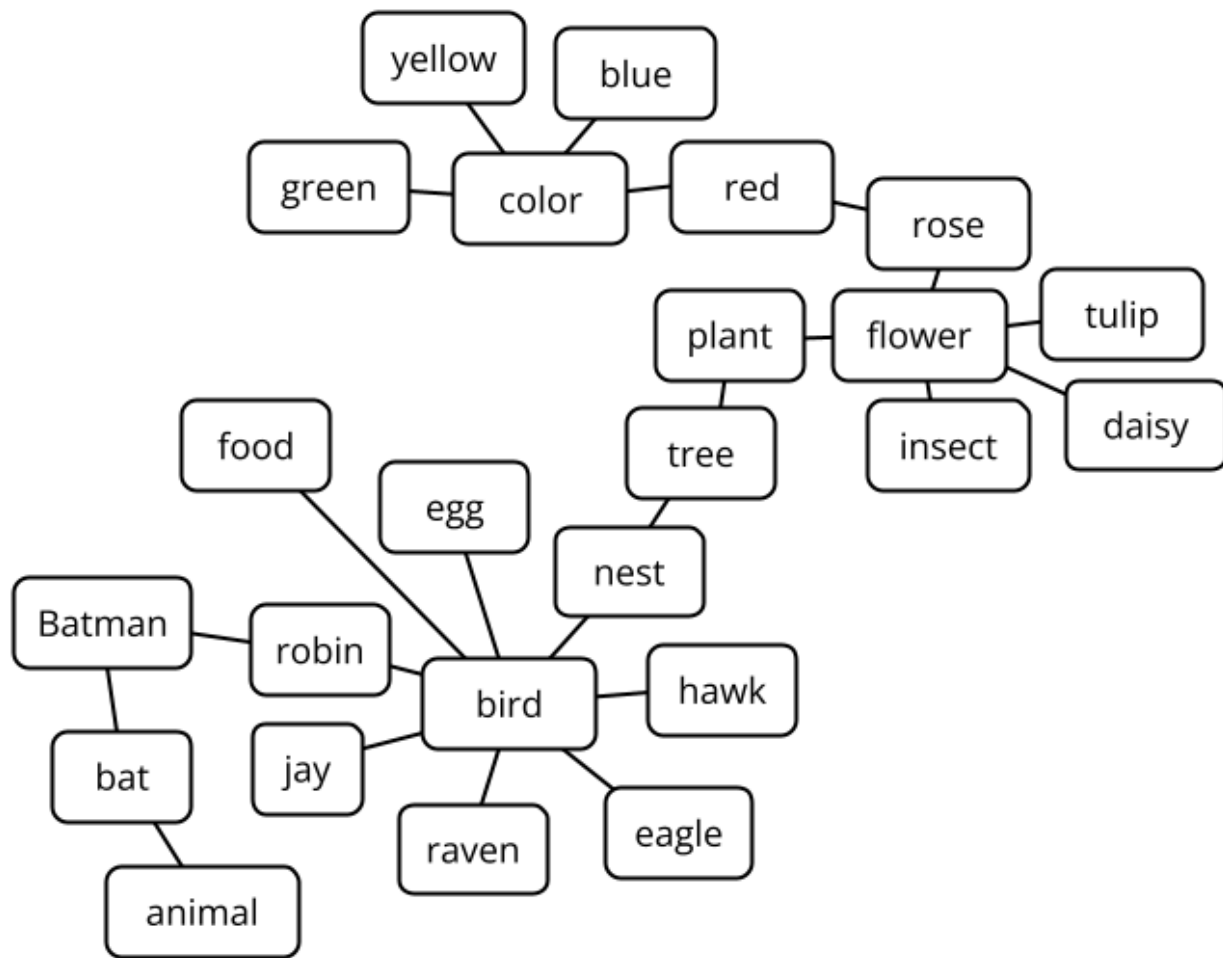


Figure 4.6 Liz's tree of associations, this time with even more items!

Feel free to peruse Liz's tree to see what her associations are. Your own may be the same, or they may be completely different. In any case, now that we have a lot more items in her associative web, we can explain the experimental procedure. To begin, please direct your attention towards the *flower* and *color* sections of the web (you'll see we got there by associating *bird* with *nest*, *nest* with *tree*, *tree* with *plant*, *plant* with *flower*, *flower* with *rose*, *rose* with *red*, and *red* with *color*). For Liz, *flower* is associated with several types of flowers, including *rose*, *daisy*, and *tulip*. *Color* is associated with several colors, including *red*, *yellow*, *green*, and *blue*.

To do the experiment, Liz will need to complete some word pairs. For example:

COLOR: Y _____

What might she fill that space in with? Likely, *yellow*. Next, up, we have another word pair:

FLOWER: R _____

Perhaps, she'll fill it in with *rose*.

You might be thinking, "Hey! This test doesn't seem very hard." And you'd be right! We're not trying to

trip Liz up here. Instead, we're measuring how long it takes her to come up with these words. Let's say her examination results look like this:

Word Pair	Liz's Answer	Reaction Time
COLOR: Y_____	yellow	1.56 s
FLOWER: R_____	rose	1.55 s
COLOR: B_____	blue	1.34 s

Now, wait a minute! Why was Liz faster to come up with *blue* than she was with *yellow*? Indeed, Liz's reaction time to name a color will change depending on how much time has passed since she last thought of a color.

Number of Distractors	Looks like...	Reaction Time to say “Blue”
0	COLOR: Y_____ COLOR: _____ B_____	1.22 s
1	COLOR: Y_____ FLOWER: _____ R_____ COLOR: _____ B_____	1.34 s
2	COLOR: Y_____ FLOWER: _____ R_____ BIRD: H_____ COLOR: _____ B_____	1.48 s
3	COLOR: Y_____ FLOWER: _____ R_____ BIRD: H_____ ANIMAL: _____ W_____ COLOR: _____ B_____	1.56 s

If Liz answers two *color* questions in a row, she’s very fast at the second *color* question. If she answers two *color* questions with one *flower* question in between, she’s a little slower. If there are two questions in between, she is slower still. The spreading activation model can explain this! This is where the “spreading activation” term comes into play. The idea of this model is not just that your concepts are stored in a web, but also that when you think of one concept, it makes you think of all the other things you associate with that concept. In other words, it “activates” the concepts around it, and that activation “spreads.” So, thinking of *bird* will make Liz think of things she associates with birds, and thinking of *color* will make her think of colors. By the time the second *color* question rolls around, Liz is already thinking about colors, so she is faster at coming up with a color. You can see a visual representation of this in Figure 4.7.

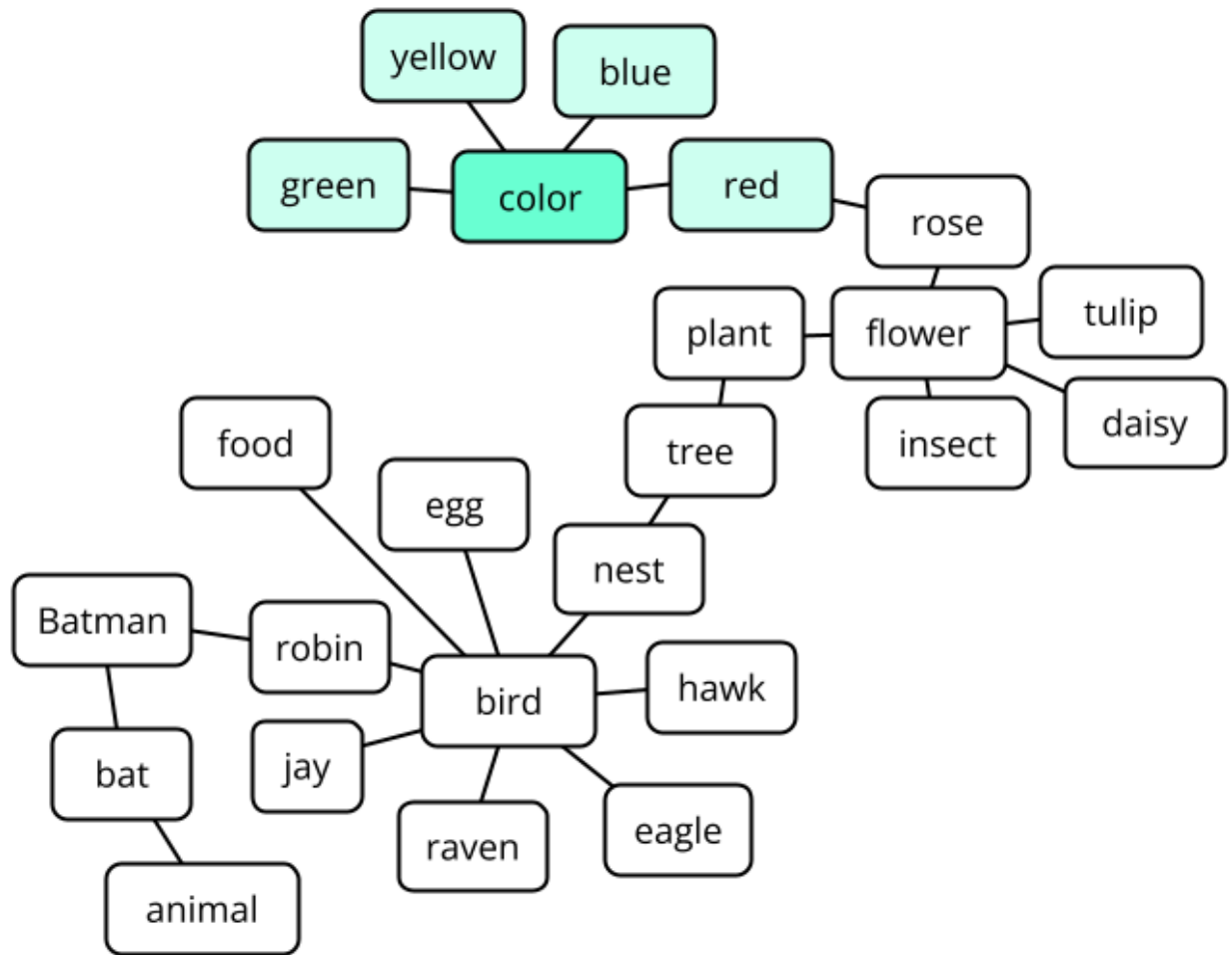


Figure 4.7 Activation “spreads” out from color after Liz thinks of it, making green, yellow, blue, and red easier to think of.

This model—that our concept knowledge is just a giant network of association—is fantastic at explaining a wide range of things, so much so that association is one of the core themes of this textbook. More than anything else, your mind runs on association. Why? One answer is neurons!

Concepts as Networks of Neurons

A neuron is a cell of the nervous system whose primary job is to communicate. As a communicator, it has a receiving end, a central processor, and a transmitting end. You can see a diagram of some neurons communicating in Figure 4.8.

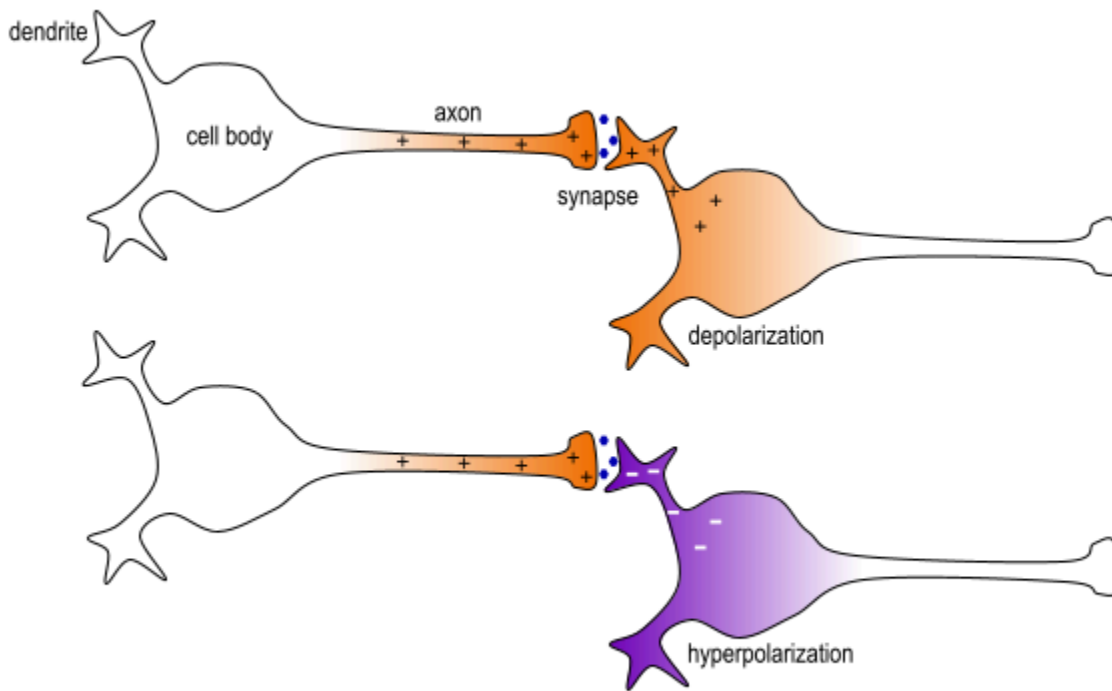


Figure 4.8 Upper panel: a neuron sends an excitatory signal. Lower panel: a neuron sends an inhibitory signal.

The receiving end of a neuron is called the **dendrite**. Here, other neurons (or other stimuli) send signals to the neuron. These signals can take many forms: neurotransmitters, other chemicals, electrical changes, or even physical touch. Often, information that stimulation has occurred will travel down the neuron, arriving at the **cell body**. The **cell body**, or *soma*, is the central processor of the neuron. In other words, it's the regular “animal cell” part of the neuron. If you have learned about cells before, you may be familiar with terms like *mitochondria*, *nucleus*, and *endoplasmic reticulum*. These are the sorts of things you will find inside the cell body.

When a neuron is firing more action potentials than average, we say that it is *excited*. When a neuron is firing fewer action potentials than average, we say that it is *inhibited*. You can examine a common way we represent this visually in Figure 4.9.

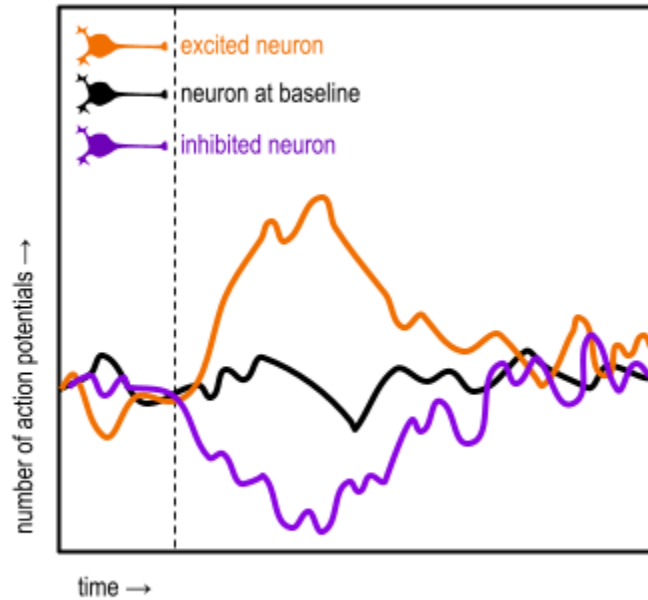


Figure 4.9 Baseline neuron (gray line), excited neuron (orange line), inhibited neuron (purple line). Dotted vertical line represents the time at which the neuron receives a signal.

One thing neurons are very good at is *associating*. When two things happen together, the connection grows stronger between the neurons that represent those things. When you're born, you don't know any concepts (or at least, you can't articulate any concepts through language). As you learn words, however, concepts form in your mind – first haphazardly, perhaps with errors (look at all those chickens), but they firm up over time. When you first see an apple, you might hear someone say the phrase, “Hey, can I have that apple?” Your brain then logs that the sound *apple* is associated with a round, red object. When you next see an apple, someone might say, “This is way better than an orange!” Your brain could then mistakenly associate that round, red object with the sound *orange*.

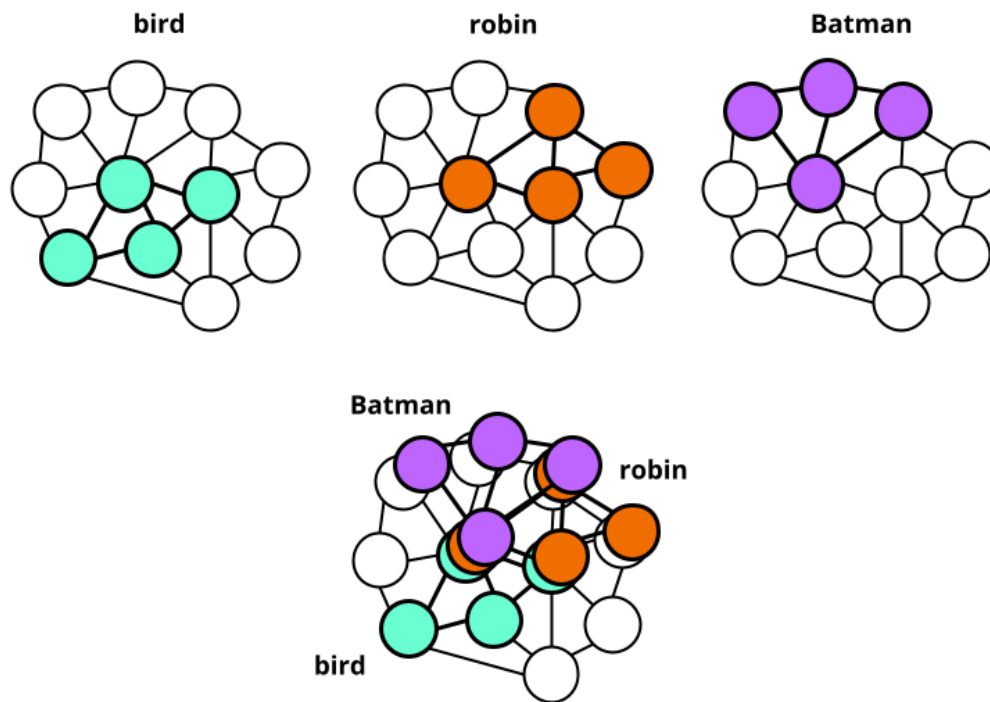


Figure 4.10
Concepts activate populations of neurons. Related concepts activate populations of neurons that overlap.

Eventually, however, you will hear the sound “apple” while looking at an apple way more often than anything else, and your brain will learn the concept *apple*, because the same neurons are firing every time you think about an apple. You can think of a concept as a pattern of neurons firing. If we add back in the idea of spreading activation, we can also conclude that when you think *apple*, you’re also activating the neural patterns associated with *red*, *sweet*, and so on. Related concepts will activate overlapping populations of neurons, like in Figure 4.10.

We’ve covered what your understanding of concepts *looks* like. In the next chapter, we’ll talk about what it *is*.

References

- Collins, A. M., & Quillian, M. R. (1969). Retrieval time from semantic memory. *Journal of verbal learning and verbal behavior*, 8(2), 240-247.
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5.

IMPLICIT MEMORY

Introduction

This book is supposed to be about *Memory* and Cognition. It's Chapter 5, and we haven't talked about memory yet! What gives?

Actually, we *have* been talking about memory! We just haven't been calling it that. We've been calling it *concepts*. Your knowledge of what things are is an example of one of your memory systems and the subject of this chapter: **implicit memory**. Implicit memory is unconscious: it's everything you know, and know how to do.

Association

Because implicit memory is everything you know, it includes all the concepts we've been discussing so far, like *bird* and *color* and *plant*. With that knowledge comes all the associative links in between those concepts. Recall our discussion of spreading activation as both a mental and neural function. The formal name for this process of memory is **priming**.

Priming is a term for what happens when thinking of one thing makes it easier for you to think of another thing. For example, consider the following list of words:

- **Strawberry**
- **Blood**
- **Ruby**
- **Tomato**

Now...think of a color!

Any color at all!

Okay. You might have thought to yourself, *I'll be clever. She obviously wants me to think of red. So, I won't. I'll think of blue.*

And that's fine! But odds are, the idea of *red* at least popped into your mind, unbidden, before I even asked you explicitly to pick a color. That is, in a nutshell, *priming*. Thinking of red things makes you think

of *red*. See the example below for an illustration of how thinking of the color white and the word silk makes a lot of people automatically think of milk.



Implicit associative memories are created through many avenues. The simplest method is repetition. If every day, you see an apple, and every day, that apple is red, you will associate *apple* with *red*. But there are other forms of associative learning; for example, **classical conditioning** – in which you associate something neutral with something meaningful, until the originally neutral thing ends up having that same meaning for you as well. You might be conditioned, for example, to feel excited when you hear the theme song of a favorite TV show, to feel hungry when you smell hot food on a grill, or even to flinch when you see something associated with someone you don't like. These are all things you don't need to consciously think about to know – which makes them implicit memories.

Procedural memory

Another aspect of implicit memory is **procedural memory** – the things you know *how* to do. And there are quite a lot of things that you know how to do! If you are reading this book, you know at least one language. You also know how to move yourself around, eat, drink, and sleep. You might add additional skills to your portfolio over time: some people will learn to cook or play basketball, others to knit or play the cello, still others to ride a bicycle, drive a car, or even steer a boat. All these things that you know how to do are part of your procedural memory. While you might initially learn these things **explicitly** – that is, by consciously going through the steps, once you’ve learned it, you usually don’t need to consciously go through the steps anymore. For example, when I learned to drive, I had to explicitly think, *okay, the accelerator is on the right, and the brake is on the left*. Now, that information is present automatically, making it a procedural memory.

Implicit Memory in the Brain

The Basal Ganglia

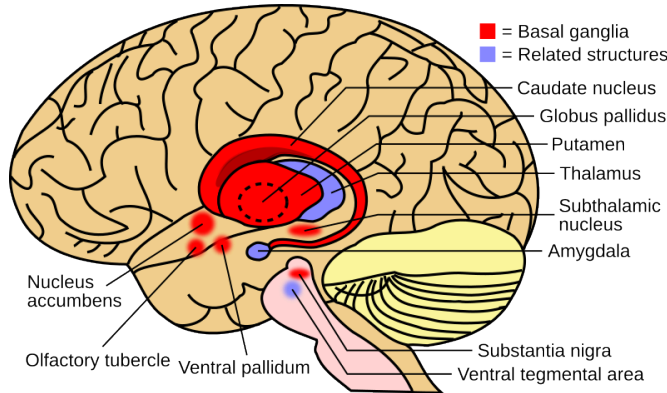


Figure 5.2 The basal ganglia and related structures, from [Wikimedia Commons](#).

The **basal ganglia** is a set of structures deep near the center of your brain primarily known for regulating motor functions, but they are also central to many habit and reward-related behaviors. Much of the signaling in the basal ganglia involves the neurotransmitter **dopamine**.

These functions make the basal ganglia an important spot for your procedural memory. You can think of procedural memories a bit like habits; procedural skills are often learned through repetition. Damage to the basal ganglia can make it harder to learn things through practice or pattern recognition.

The Hippocampus

Please try the following exercise, if you can:

- Close your eyes!
- Point to the nearest door!
- Open your eyes and check – were you correct?

Odds are that (if you were somewhere indoors), you were correct. Why? Did you try to memorize where the door was when you entered this current room? Perhaps, but likely not. And yet, you usually know where all the doors are around you. In fact, you know a lot about what's around you and how you got there. Your brain is actually tracking your location all the time, building a little map of the world around you. Your **hippocampus** is one of the structures primarily responsible for this process.

The hippocampus accomplishes this incredible feat in an incredibly organized way. Consider Tolly the rat – you can observe him again in Figure 5.4, exploring a rectangular room. The figure's top panel shows Tolly's path: he's cautious, staying around the edges, but he ultimately can't resist visiting the cheese in the open space before exiting.

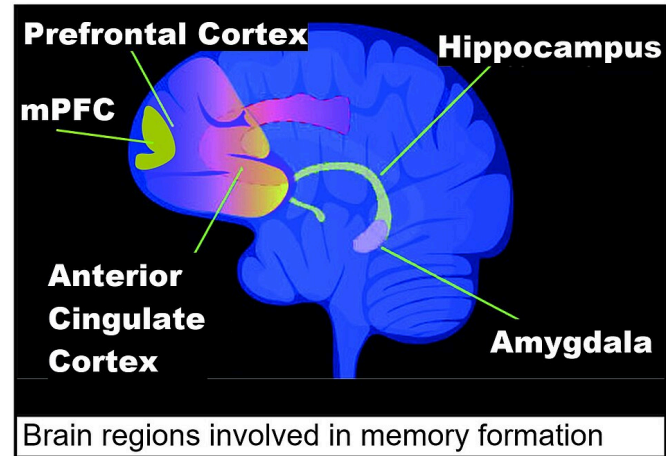


Figure 5.3 The hippocampus is located towards the center of the brain, from Wikimedia Commons

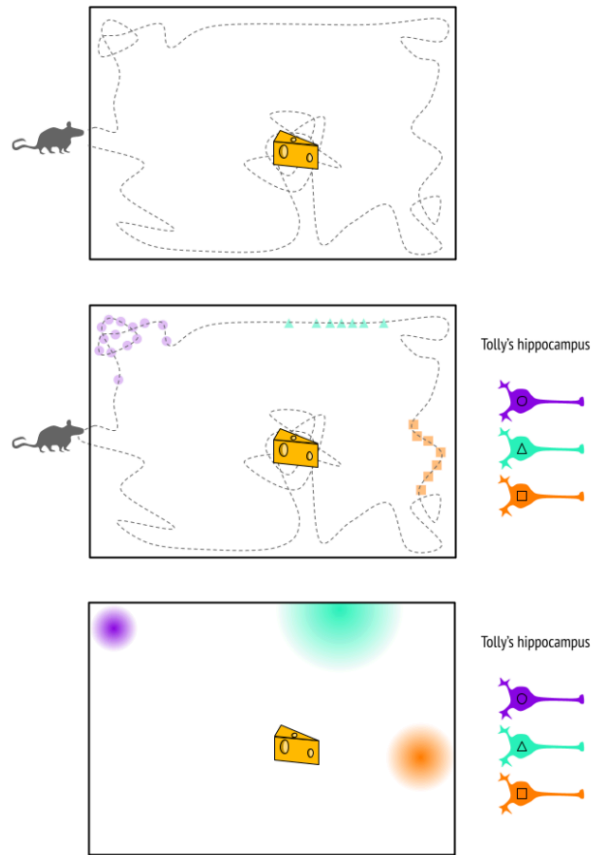


Figure 5.4 Hippocampal place cells prefer specific locations.

The middle panel is where things start to get interesting. Here we have three neurons from Tolly's hippocampus to observe: a purple neuron, a green neuron, and an orange neuron. Each neuron only fires an action potential when Tolly is in a certain area of the room. When Tolly is in the upper left of the room, the purple neuron fires (each action potential is indicated by a purple circle). When Tolly is in the top right-center of the room, the green neuron fires (action potentials indicated by green triangles). Tolly is in the lower-central right side of the room, the orange neuron fires action potentials (indicated by the orange rectangles). These neurons are known as **place cells**¹. The colored zones in the lowest panel of Figure 5.4 represent each place cell's favorite area of the room; whenever Tolly enters each part of the room, the place cell that likes that part the best fires an action potential! Tolly's brain does this for every space he enters – it keeps track of where Tolly is in space. This discovery of place cells (and subsequent work that fleshed out the brain's mental maps even more²) actually earned its discoverers a Nobel Prize in 2014!

Chapter 5 References

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Moser, E. I., Kropff, E., & Moser, M. B. (2008). Place cells, grid cells, and the brain's spatial representation system. *Annu. Rev. Neurosci.*, 31(1), 69-89.

6.

EXPLICIT MEMORY

Introduction

The term *explicit memory* refers to knowledge you consciously know. You can think it in words or images, and state it out loud. In this chapter, we'll discuss different types of explicit memory, as well as some common ways in which these memories can be inaccurate.

Episodic and Semantic Memory

We divide episodic memory into a few different categories. One such category is **episodic memory**—your memory of events. This includes things like your memory of a social gathering you attended, a sporting event you participated in, or a show you watched. Most of these memories are going to have you in them, since you're the one remembering them, but we sometimes refer to specific episodic memories that are about you as **autobiographical memory**.

Episodic memory is your memory of things that happen, but what about your memory of the facts and details that fill in those events? That is your **semantic memory**—your memory of *things*—facts, pieces of information, and concepts. One can actually represent your semantic memory as a web or network of concepts, connected by association...Surprise! We actually spent a large chunk of the first part of this book already discussing semantic memory, we just didn't call it that.

While your memory of your sixteenth birthday party might be an episodic memory, your knowledge of your age and the name of the town you were in at the time are pieces of semantic memory. There can be overlap between the two; don't worry too much about trying to draw a clear line between semantic and episodic memory. For now, let's dig a little deeper into semantic memory. We already know a good way to represent how it's organized in your brain. But how do we access it? What is it like to use it?

A really simple way to examine your semantic memory is to test your ability to memorize a list of words. For this exercise, you'll need something to write on, and something to cover up the list once you've finished looking at it.

Got them? Good! Here are your instructions:

- Next, you'll see a list of 20 random words, and your goal is to memorize them.

- Read them aloud, slowly, one at a time, once all the way through.
- Then, do it again!
- When you are done reading the list twice, cover them up, and count backwards from 10 to 0!
- Finally, write down as many words as you can recall.

Ready to go? Here is your list!

- Income
- Office
- Drawing
- Anxiety
- Medicine
- Studio
- Candidate
- Family
- Wife
- Growth
- Grocery
- Height
- Indication
- Contract
- Two
- Orange
- Hat
- Obligation
- Importance
- Drama

Congratulations! In the spirit of fairness, I did the exercise as well. Here's what I came up with:

- Income
- Orange
- Hat
- Drama
- Obligation
- Science

I remembered 5 words from the list (and one that snuck in there that wasn't on the list at all). Consider how you did! Specifically, consider the following questions:

- Did you remember *income* (the first word?)
- Did you remember *drama* (the last word?)

If you remembered the first or last word, you did something most humans do! In fact, you can graph the general pattern of how people tend to remember items from lists like this:

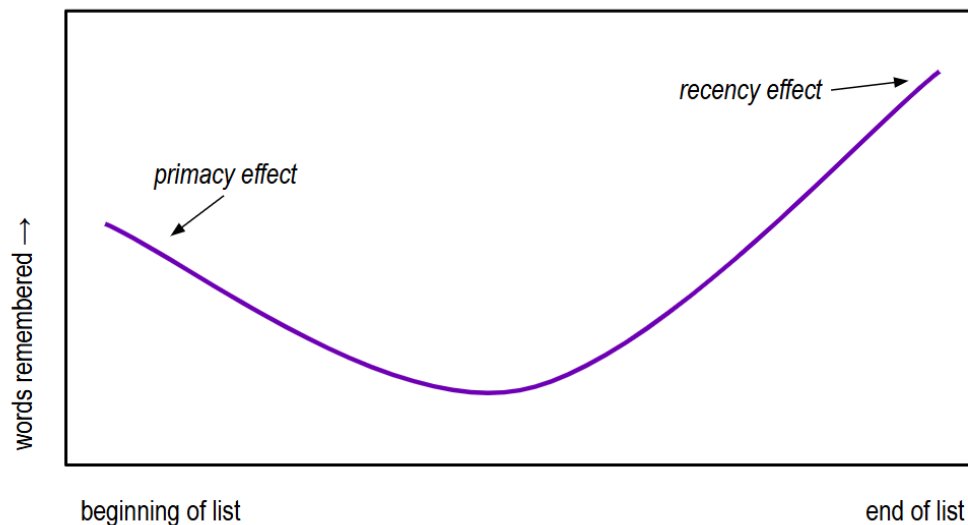


Figure 6.1 Serial position effects.

There are a few things going on here, and they're all part of an umbrella of phenomena called *serial position effects*. First, we often remember the first word in a list. This is called the **primacy effect**. We also tend to remember things that *just* happened. This is called the **recency effect**. As for things in the middle of the list...they often get neglected.

Do our brains really care about beginnings and ends? Is it a meaningless artifact of how neurons work? There are many models out there that try to explain how or why these effects occur, but the important thing for us to know is that they happen consistently: all semantic memory is not equal! How you take in the information matters.

False Memories

Arguably, one of the most important things to know about semantic memories is that many of them are false! False memories are extremely common and a natural side-effect of how memories are formed, stored, and remembered.

The War of the Ghosts

In the early 1900s, a British psychologist named Frederic Bartlett performed a very influential experiment illuminating how memories change as they are stored in our brains. This study is known by the title of the story that Bartlett asked subjects to read and remember: *The War of the Ghosts*. Now, *The War of the Ghosts* is allegedly an Indigenous American story and the fact that it had a cultural context that was unfamiliar to the subjects of the experiment was important. However, I can find no evidence that it came from anywhere except Bartlett himself, so I'm a little concerned that it has nothing at all to do with any specific Indigenous people, and that Bartlett just made it up. So, let's set that aside and work through an example story that I generated using AI, with a few of my own tweaks. Don't worry, it will serve the same purpose.

We'll call this one *The Man in the Mansion*. Please read it now:

Once upon a time, there was an old mansion on a hilltop. Fred was walking through the mansion, trying to decide if he wanted to buy it. It was dusty and cold inside the house, but Fred didn't mind. He knew dust was a sign of good fortune. Fred walked up the staircase, which turned at a right angle, and leaned over the balcony to look down over the entrance hall. He rapped on the railing with his knuckles to make sure the balcony was secure. It rattled but didn't break. Satisfied, he snapped three times with his index finger and thumb. As the clock struck eight, it suddenly became very dark in the house. Fred gasped. Two figures erupted from the house riding headless horses. Fred yelped in surprise! The two figures ignored him. They jumped towards each other and began to punch each other. Fred ran downstairs, making sure to turn left, and stood between them. He begged them to stop fighting each other. They were going to damage the house! He held out his arms wide...but they passed right through him. They were ghosts! Fred suddenly felt very warm, and his fingertips were blue. He looked up at the ghost to his right. Fred asked the ghost what was going to happen to him, and the ghost to his left laughed and did not answer. Fred looked up at the ghost to his left and asked the ghost what was going to happen to him, and the ghost to his right laughed and did not answer. Fred looked up at the ghost to his right and asked him what was happening, and the ghost to his left laughed and told him it had already happened. Fred looked down at his body and saw that it was transparent. He was a ghost too!

Perhaps it is not a classic of modern literature, but it should do for our purposes.

Now that you've been distracted for a few seconds, please recall every detail in the story that you can, to the best of your ability – as if you were telling it to a friend.

Done? Great! Here are some things that might be true regarding your recollection of the story:

- Your recollection is shorter than the original story
- Your recollection might not be 100% accurate
- Your recollection is unique to you

The Man in the Mansion was 305 words. Your recollection may or may not have been less than that, but odds are that even if you're a particularly wordy person, you omitted a few details from the original. And that is by design.

Your brain is not a machine that makes copies of things. That is a good thing! It would be a huge waste of energy. Most of the time, we don't need every detail of something. When you attend a baseball game, you don't remember every single pitch; you just remember the exciting bits. When you see a movie, you remember the plot, and some specific things that stand out, but not every single second of the runtime.

Suppose you've saved a document on a computer. When you open the document, the words are exactly the same as they were when you left them—the computer saved the file with every detail intact. This is not what your brain does! When you “save” a memory in your brain, what is saved is a set of highlights, associations, and feelings. When you “re-open” the document to recall the memory, your brain brings up those flashes and fills in the rest on the spot.

In short, memory is what we call **constructive**. It does not mimic things; it creates them. The consequence of this is that every piece of information you've ever received from another human has a bit of them in it!

Creating False Memories

In the 2010 film *Inception*, the protagonists attempt to put ideas into the minds of others by travelling *into* their minds. The process is depicted as extremely risky and dangerous. However, “inception” is a lot easier in real life.

If a psychologist came up to you and told you that when you were six years old, you went on a hot air balloon ride with your family, would you believe them? Perhaps, but if you had no memory of this event, you likely won't remember it. And you shouldn't, because it never happened. The psychologist made it up.

But what if the psychologist had a picture of you and your family, there on the hot air balloon? What if they also reminded you of several stories that you do remember? What if you were asked to describe the memory to the best of your ability, repeatedly over several days?

Under those circumstances, people can sometimes generate false memories! Even if you never did go on a hot air balloon ride when you were six, you might come away from the interaction believing that you did.

This is a common format for a psychological experiment. In the table below, you can see several examples

of psychology experiments that successfully convinced a portion of their subjects that they remembered something that in fact did not occur.

Experimenters convinced subjects...	By...	Citation
They took a kid on a hot air balloon ride	Showing them a doctored picture	Wade et al., 2002
They pranked a teacher	Telling them true stories alongside the fake story	Lindsay et al., 2004
They saw broken glass in a car crash scene	Asking them if they saw “the” broken glass	Loftus & Palmer, 1974
They spilled juice on someone at a wedding	Telling them true stories alongside the fake story	Hyman et al., 1995

All of these studies share a few important features. Many of these studies incorporated repetition. If you go up to someone and simply tell them “you went on a hot air balloon when you were six, don’t you remember?” it might not work. But if you go over the “memory” many times, over many different sessions, the person might start to form one.

Moreover, the studies above take advantage of subjects’ trust. Sometimes, they simply rely on the general faith people have in scientists, but often, they go even further. They provide subjects with falsified evidence of the event they are trying to convince them happened or hide the lie amongst truths and real details. Either way, it’s much easier to believe something when you assume the source is trustworthy and have legitimate evidence to back that up. In the following example, Dr. Gotcha fools Sunny by triggering real memories before probing for a false one and wearing his Science Goggles to appear more authoritative.



Remember when you went on a cruise at age 10?

I sure do!



What about when you went on a train?

Yup!



And when you went on a blimp?

Hmmmmm...no...



Wait, look at this picture of you on a blimp!

Oh yeah! The blimp ride!



Gotcha! You never went on a blimp ride.

Can you imagine if I filled this textbook with lies? Don't worry, I didn't. But are you actually going to follow up on all my citations? What about everything I stated as fact but didn't cite? Probably not —that would be a huge waste of time and defeat the point of the textbook. But if you don't follow up on and fact-check every single thing I've said in this book, you're taking a mental shortcut of sorts. You're assuming that this is not a book of lies. Our brains are not wired to be constantly trying to catch lies. This makes life easier, but it also leaves our memories vulnerable.

Eyewitness Memory

Imagine you're in a courtroom for a car mishap, and it matters whether you had been speeding or not. If someone asks a witness how fast they thought you were going, you'd hope their answer would be accurate. Now, we already know that memory isn't always accurate, but what if on top of that, the *way* the lawyer asked the question actually influenced the witness's answer?

In the 1970s, a psychologist named Elizabeth Loftus asked people to watch a video of a car accident. Afterwards, she asked them to tell her how fast they thought the car had been going. But there was a catch! Different subjects got slightly different worded questions. You can see the effect of these different words in Figure 6.3.

When deciding how fast the car was going when it *contacted* the mailbox, subjects reported its speed to be around 30 mph. When deciding how fast it was going when it *smashed* the mailbox, they reported its speed to be around 40 mph. The wording of the question alone was strong enough to influence the subjects' memories! And we already know why—it's because memory is constructive. When you think back to a thing you saw, you aren't "replaying" it in your mind – you're *re-building* it. And sometimes, irrelevant or noticeable information around you creeps into that reconstruction. If someone asks you to re-build a memory of a "smash," you're going to construct that memory with a little more intensity than you would if they asked you about a "contact."

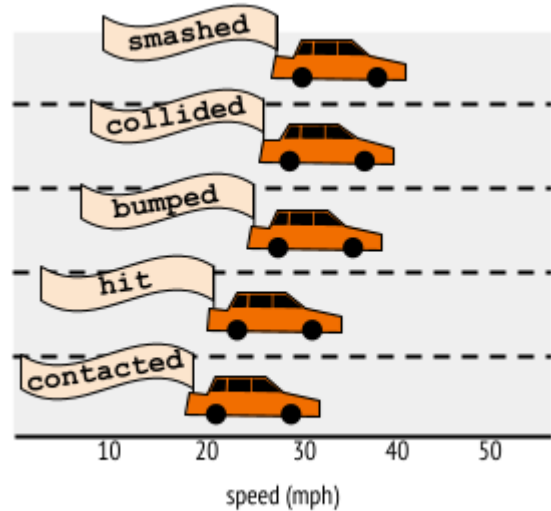


Figure 6.3 Subjects remembered the car going faster if experimenters asked for the speed using a more intense word for the collision.

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7.

ACCESSING MEMORY

Introduction

So far, when we have been discussing memory, we have primarily discussed **long-term memory**. But there are other kinds of memory, and they have to do with how we get everything into long-term memory in the first place. So, how does information enter your mind?

Sensory Memory

There are only two sources of information available to you. Information from within yourself, and information from outside yourself. The within-yourself stuff, well, that's for a future chapter. But the outside-yourself stuff is...your senses! That's it. There is no other way to get information from "out there" into you. And contrary to popular opinion, you do have more than five senses. The "big five" are known as sight, smell, hearing, touch, and taste. But you also have pain, temperature, balance, acceleration, body position...and most of the time, most of these senses are all giving you information at once.

For example, as I write this, I am currently wearing a shirt, pants, and socks. In other words, maybe 80% of my skin is currently touching some kind of fabric. That means that 80% of my skin cells are all constantly shouting at my brain, saying "I'm touching something! I'm touching something!" once every few milliseconds. But am I paying attention to any of that? Of course not! In fact, most of the sensory information I get at any given millisecond, I immediately throw away. The feeling of socks on my feet is important at exactly three moments: 1) when I put them on, 2) when I take them off, and 3) if something interesting happens to my foot. For example, one of my cats may brush up against my foot. In that case, I am aware that there is a sock between my skin and his fur. Other than those three scenarios though, I don't need to be constantly updated that my sock is touching my foot. I ignore it.

This is the experience of **sensory memory**—that millisecond-scale awareness you have of everything you sense—you hang on for just enough time to comprehend the world around you, and then you dump most of it and move on. To understand the second half of this sentence, you need to remember the first half. To understand the right side of a room, you need to remember what the left side looked like. This form of memory is crucial in order for you to have a coherent experience of your world, but otherwise, it's fleeting, and there is no reason to save most of it for future use.

But the miniscule fraction of information you *do* save becomes...

Short-Term Memory

If you want to save something in your awareness for more than a few hundred milliseconds, you turn it into **short-term memory**. Short-term memory is your ability to hold on to something in your mind for a limited amount of time. We'll now break down the two rather vague terms in that definition: "something" and "limited amount of time."

Items in Short Term Memory

Your short-term memory can contain any kind of information—the sound of a line of a song, the taste of a piece of cake you just swallowed, the contents of a list you're trying to memorize, and more. To get a sense of what your short-term memory is like, we can start with an exercise: please take about 30 seconds to try to memorize the following numbers:

3 2 7 1 4 1 7 2 3 3

If you're like me, you might have repeated these numbers "verbally" in your head until they stuck: an internal monologue that sounds like "three, two, seven, one, four..." and so on. This is an example of the auditory component of short-term memory. If hearing is part of your life, it's usually one of the ways in which you try to keep things in your head. Formally, we call this process **rehearsal**.

You might have found it a challenge to memorize all 10 of those digits in 30 seconds, or at least that it took some mental effort. Holding something in short-term memory can feel like trying to hold water in your hands—you have to try very hard to keep it there, but you can't hold it forever.

Limits to Short Term Memory

How long, exactly, can you hold something in your mind before you either forget it, or you remember it forever? While the general time span for short-term memory is anywhere from a few seconds to a few minutes, a different way to describe the limits of your short-term memory is not in time but in pieces of information.

Consider the 10 digits I asked you to memorize above. That was 10 pieces of information. Ten is on the high end for "numbers of things it's reasonable to ask someone to remember." If you think about various strings of digits you may use in your life—ID numbers, PIN codes, license plates—the ones that people most often assume you know off the top of your head tend to be shorter than 10 digits. A 10-digit PIN would be much more secure than a four-digit one, but I am much more likely to remember four numbers

than ten! In fact, a scientist named George Miller attempted to formalize this phenomenon in the 1950s, landing on the “magic number” seven...plus or minus two¹. In other words, people can remember at most about 5-9 individual items before it all becomes too much.

Have you ever managed to easily remember more than nine things at once? It’s very likely that you have. There are several glaring exceptions to the claim I made above—that most numbers people expect us to have memorized are short—the most glaring of which might be...phone numbers! You might be thinking, *I know a lot of phone numbers, and they are all ten digits long! What gives?* Ah, but there is something very special about phone numbers and the way they are presented to us, something strange and magical, with a very whimsical name...it’s called a *chunk*.

Here are those ten digits from above again, but this time, they are *chunked*:

3 2 7 – 1 4 1 – 7 2 3 3

Hey, it looks a bit like a phone number now! Indeed, splitting items into groups is a very effective way of tricking our brains into accepting much more information. When you try to memorize a number like this, instead of saying to yourself “three, two, seven, one, four...” you might say something like “threetwoseven...onefourone...seventwothreethree.” You combine ten individual items into three groups, fooling your brain into thinking you’re memorizing three things, not ten. This strategy is very effective in increasing the number of things we can remember – though you will ultimately hit a limit of *chunks* you can hold in your short-term memory too. If you find yourself remembering long strings of information for longer than a few seconds or minutes, congratulations! You’ve successfully transferred something into *long-term memory*, which is everything you know and remember, and, well, we’ve already talked about it! But there is one more kind of memory we haven’t addressed yet that might help glue all these pieces of memory together. Let’s look at some memory models and introduce **working memory** to the party.

Working Memory

So far, we have your sensory memory, short-term memory, and long-term memory. You can represent these in a model like in Figure 7.1.

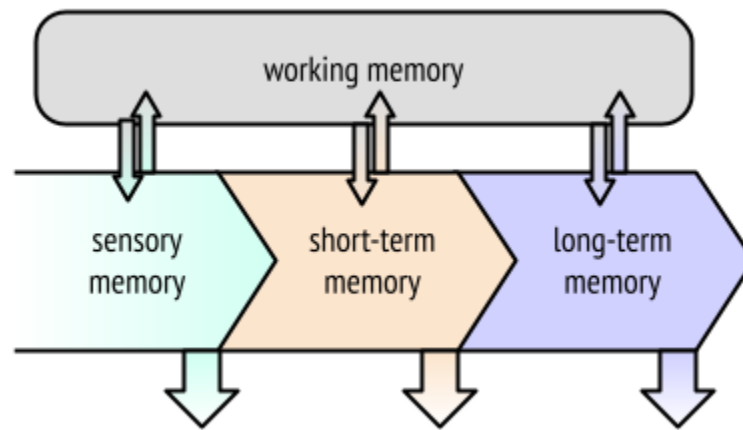


Figure 7.1 A model of sensory, short-term, long-term, and working memory

In this model, sensory memory feeds into short-term memory, and short-term memory feeds into long-term memory. Importantly, loss of memory is possible at each of these stages. Sensory memory doesn't make it to short-term memory unless you pay attention to it, so inattention is what causes memories to leak out of sensory memory. Short-term memory doesn't make it to long-term memory without effort, so lack of effort causes memories to leak out of short-term memory. As for long-term memory loss...we will discuss that in the next chapter.

Above these three forms of memory is a block that spans all three called **working memory**. *Working memory* is a term for when you are consciously accessing memories and manipulating them. You're not just experiencing something or holding it in your mind; rather, you're expending some effort to accomplish a task. An example will help make this clear. Consider the following series of letters:

L H B S Q K

Now try the following exercise!

- Memorize the letters, close your eyes, and recite them
- Now, close your eyes and recite them...in alphabetical order!

You may have noticed that the second task took more effort than the first task. That's because the first task was only asking you to use your short-term memory, but the second was triggering your working memory. To recite the letters from memory in alphabetical order, you need to dip into not only your short-term memory but also your sensory and long-term memory! The process might look something like this:

- Use short-term memory to rehearse the letters: "L H B, S Q K, L H B, S Q K, L H B..."
- Decide to figure out which letter of those six is the earliest in the alphabet
- While the rehearsal is still running, open up long-term memory to access the alphabet, and determine

that “B” is first

- Sensory memory helps keep “B” alive as you transition from long-term to short-term
- Hold both the rehearsal “L H B, S Q K,” *and* the fact that B is first in your mind
- Open up your knowledge of the alphabet again and repeat!

This is what it is like to use your working memory. More broadly, this is what it is like to think with effort and planning – a task very similar to something we’ve already discussed: **executive control**. Indeed, working memory and executive control have a lot of overlap. Which term we use to refer to the process of “figuring out what to do next based on current information and past knowledge” depends largely on context.

Explicit Memory in the Brain

The Hippocampus

We’ve already seen the hippocampus (in **Chapter 5**, when we discussed place cells). Spatial memory is not the only function of the hippocampus! It plays an interesting role in memory storage as well.

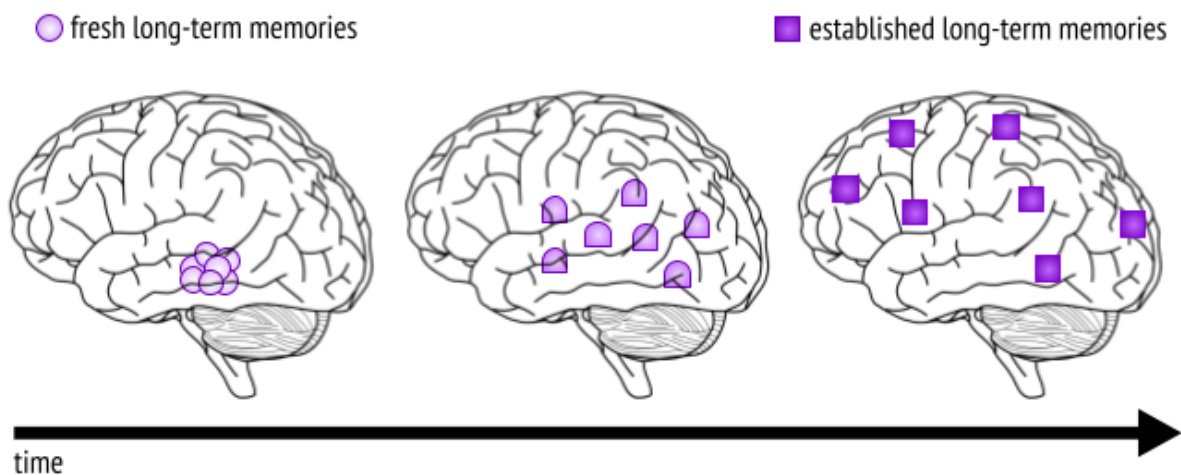


Figure 7.2 Explicit memories begin in the hippocampus and are distributed throughout the cortex over time.

Long-term memory is not all the same. You likely experience this fact all the time; some things come to mind much easier than others. Memory is also more or less vulnerable to forgetting and loss under different circumstances, because memory is stored differently in different circumstances! Within long-term memory, there is a difference between *fresh* explicit memories (memories you’ve formed in the last few hours or days) and *long-term* explicit memories (memories that stick around over time). **Figure 7.2** represents the journey of long-term

explicit memories into permanent(ish) storage. The fresh, new long-term memories are kept in the hippocampus. But over time, they are distributed all over the brain.

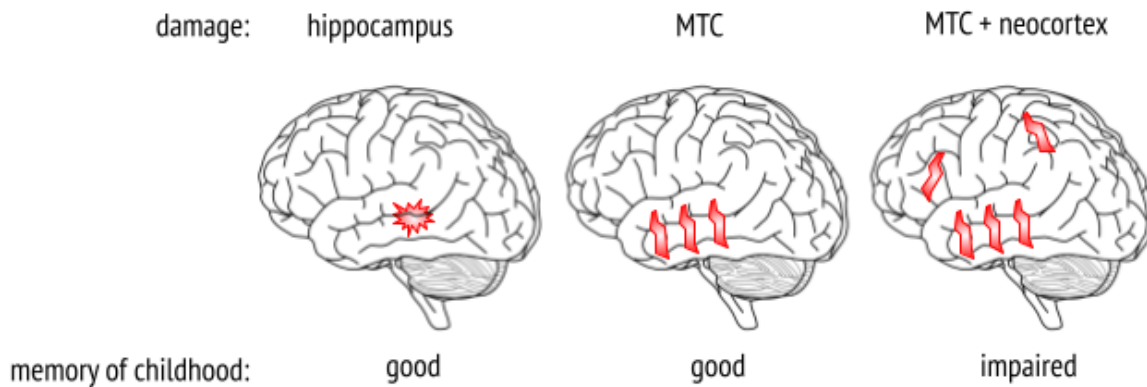


Figure 7.3 Recall of childhood memories with different areas of brain damage.

This is one reason why it's hard to lose very old, very established memories, and nearly impossible to have your personality “wiped” through memory loss. In order to lose *all* your explicit memories, you'd need to lose a lot of brain mass. Figure 7.3 shows a comparison of autobiographical memory² between people with various types of brain damage. Because autobiographical memory stretches back a long time and is reinforced frequently, it's hard to disrupt it: people in this study with damage to their hippocampus demonstrated an intact autobiographical memory. People with damage to their medial temporal cortex (in the temporal lobe, close to the hippocampus) also demonstrated an intact autobiographical memory. The only subjects who struggled with their autobiographical memory had damage not only to their medial temporal cortex but also to other regions of their cortex – in other words, more extensive and broad brain damage.

Memory Loss

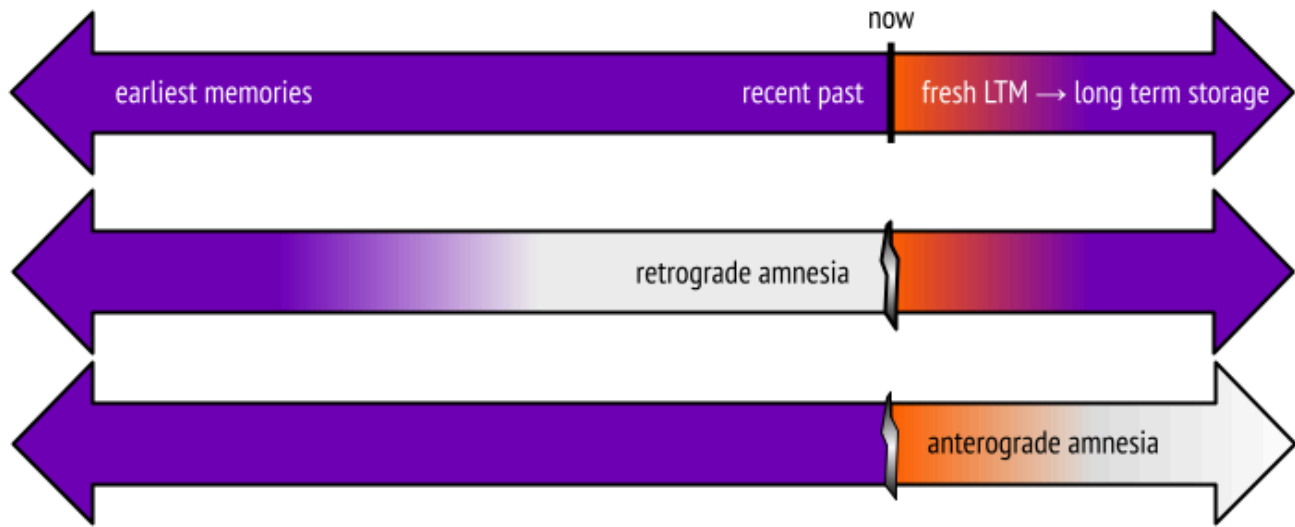


Figure 7.4 Retrograde amnesia causes loss of past explicit memories, while anterograde amnesia causes the inability to consolidate new explicit memories.

There are several different forms of memory loss. **Amnesia** is one of the best-known: in the movies, it's often a complete “wipe” of everything you know. But there are actually several different types of amnesia! We will discuss two below. They are quite different, but both involve memory loss, and both are usually triggered by some kind of brain injury (often stroke, surgery, or accident). You can compare how each works in Figure 7.4.

Retrograde Amnesia

When we think about memory loss, it's perhaps most natural to think about forgetting the past. One form of forgetting the past is **retrograde amnesia**, in which you lose explicit memories that you had before the injury occurred. For example, if you hit your head at 9 PM on your twenty-seventh birthday, you may not remember anything before 9 PM on that day. You may not even remember your whole 26th year of life. You can lose quite a lot of your past memories, but very importantly, you usually don't lose it *all*.

This is for two reasons: one, recall our discussion about how long-term explicit memories are distributed throughout your brain. While retrograde amnesia can occur from injury to a variety of brain locations, it's unlikely that one injury can ruin every single one of your memories—especially your earliest ones. Even if you can't remember the last decade of your life, you will likely still know your name and understand who you are.

The other reason is that we are focusing on explicit memories here. The term *retrograde amnesia* generally does not apply to implicit memories, and those tend to remain intact in these cases. So, even if you don't remember the last month of your life during which you learned how to ice skate, you will still know how to ice skate. You just won't remember learning it.

Anterograde Amnesia

Recall our discussion of “fresh” versus “established” long-term memories. Retrograde amnesia refers to the loss of some of these “established” memories stored throughout the brain. But what happens if you experience damage to your hippocampus, where the “fresh” long-term memories live?

Hippocampal damage can sometimes result in **anterograde amnesia**, in which you lose the ability to transfer memories into long-term storage. New explicit memories appear for you, but they don't stick around for more than a few minutes or hours. In other words, you can't learn anything new (explicitly). A few times a day, you will forget everything that happened to you so far that day. If your injury happened on August 26th, 2017, you will wake up every morning believing it is August 26th, 2017.

If this sounds difficult, it is. Living with anterograde amnesia requires a good support system—but it is not impossible. People with anterograde amnesia can still learn new things (implicitly), they just won't remember that they learned them. There is no way to “cure” amnesia, but your brain is the most flexible thing in your body. With the right kinds of support now and more research in the future, living with brain injuries that affect your cognition will only become more manageable as the years go by.

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8.

COGNITIVE SHORTCUTS

Introduction

That's quite enough about memory, memory, and more memory, don't you think? Now that we have a clear picture of how information is stored, accessed, and lost in our minds, let's return to talking about what we *do* with all that information. So far, we've covered visualizing and organizing it. What about *using* it to solve problems?

Suppose Agatha is faced with the following math problem.

$$312.757 \times 4$$

She can approach this problem in a couple of different ways – you can see them in Figure 8.1.

$\begin{array}{r} 312.757 \\ \times \quad 4 \\ \hline \end{array}$	$\begin{array}{r} 300 \text{ ish} \\ \cancel{312.757} \\ \times \quad 4 \\ \hline 3 \times 4 = 12 \\ \text{add } 00 = \sim 1200 \end{array}$	$\begin{array}{r} 1322 \\ 312.757 \\ \times \quad 4 \\ \hline 1251.028 \end{array}$
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Figure 8.1 Two ways of solving a math problem.

Agatha can either do all the math to get the exact answer (1251.028), or she can do a quick approximation in her head and get “about 1200”. Which route she takes will depend a little on why she needs the number, but the question for us to consider is more general. What kind of calculation does a computer *always* do? And what kind of calculation do our *brains* default to?

A computer, as you likely guessed, is built to perform precise calculations. A calculator will never tell you “about 1200.” It will tell you the exact answer. And while we learn in school all the steps to come to the exact answer through multiplication, that process is not automatic or natural to our brains. When given the option, our brains will almost always choose “easy and mostly accurate” over “lots of work and always accurate.” In the case of math, that sometimes means rounding numbers and ignoring precision. Other

times, it means ignoring the digits completely and simply turning the numbers into concepts, like “a lot” and “way too much.”

When our brains simplify things like this, we call it a **heuristic**: a cognitive shortcut designed to generalize about a problem, to be accurate *most* of the time instead of *all* the time. And because they are accurate *most* of the time, we don’t always notice them. But we can expose these heuristics in the rare occasions during which they do make a mistake. In the following sections, we’ll highlight some of these errors to understand what they reveal about the assumptions our brains make.

Representativeness

Let’s learn a bit more about Agatha.



Figure 8.2 Agatha’s personality.

Agatha is 37, likes reading mystery novels, doing puzzles, writing poetry, playing with her cat, and baking soft pretzels.

Is Agatha more likely to be a librarian or a saleswoman?

You might be leaning towards *librarian*. After all, Agatha likes books and quiet indoor hobbies! It seems to fit her personality a lot better than *saleswoman*. But according to a dominant cognitive scientific perspective, you’d be dead wrong.

Why? Because there are many, many more salespeople in the world than librarians. In the United States (as of around 2020) there are around 13 million salespeople, and 166 thousand librarians. That means any random person in the United States is about 78 times more likely to be a salesperson than a librarian!



Figure 8.3 The approximate ratio of salespeople to librarians in the United States.

According to traditional cognitive science, Agatha's personality is irrelevant. Much more important is something called the **base rate**: the natural amount that something occurs. Personality varies across people, but the fact remains that there are far more salespeople than librarians, so if you draw a random human out of a bag, you should always bet on that person being a salesperson rather than a librarian.

The error that people make when they assume Agatha should be a librarian is an example of the **representativeness heuristic**: we classify things based on their traits. Why do we think Agatha should be a librarian? She has traits we often associate with libraries. Why does a tomato seem like a vegetable, when it is not one? It shares a taste profile with vegetables and often goes in the same context as vegetables. Why does a dolphin seem like a fish, even though it is a mammal? Because it shares many traits with fish, like its shape and habitat. The thing is, we are usually correct when we make assumptions like this. Most fish-like things *are* fish. That's what makes this mental shortcut useful—it's correct most of the time.

Anchoring/Adjustment

Time for a pop quiz! Please answer the following questions to the best of your ability.

- How many known species of bees do you think there are in the world? For reference, there are over 300 species of monkey.
- About how many people do you think lived in the city of Westbrook, Maine, as of the 2020 census? For reference, that year about 550,000 people lived in Portland, Maine.

Take a look at your answers. How did you come up with them? Did you guess wildly? Did you use prior knowledge? Did you use my helpful reference facts?

Was your answer to #1 smaller than your answer to #2?

Odds are that even if you ignored my helpful reference facts (or intentionally tried to block them out of your mind) they may have wormed their way into your brain to influence your answers. While both are true, the reference I gave you for #1 is much smaller than the reference for #2. If your brain was nudged at all by this information, you were employing the **anchoring/adjustment heuristic**: when we have to make a guess, we often latch on to nearby and potentially irrelevant values to help us along.

Off the top of my head, I don't have a good idea of how many bee species there are, so when evaluating #1, my thought process might be something like, *"Well, I know there are a lot more insects than monkeys, so maybe there are 10,000 species of bee."* For #2, my thought process would be, *"Should I know Westbrook, Maine? Is that a famous city? Maine is a beautiful state. But does it have a lot of people? Portland is probably its most populous city. Westbrook must be much smaller. 500,000 people smaller. Let's say it has 50,000 people."*

So my final answers would be:

- 10,000
- 50,000

In fact, both answers are about the same: around 20,000.

You can see the result of a very simple example of anchoring/adjustment in the example below, which shows what happened when experimenters asked subjects to guess the answer to a math problem. In this example, subjects fixated on the first few numbers they saw: if the first couple numbers were small, they ended up guessing a smaller number for the answer. If the first couple numbers were larger, they guessed a larger number.



Hey Liz, what's $8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$?



2250?



Hey Sunny, what's $1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8$



512?

You likely use the anchoring/adjustment heuristic all the time, because it saves you a large amount of mental effort! At the grocery store, how do you know if a bottle of lemonade is too expensive? Do you do a lot of mental math to figure out the optimal price of lemonade, or do you just compare it to the price of a different juice next to it, or the price at another store? It's usually a lot easier to find a point of reference when we make judgments about value, and this is usually effective! Using the price of other juices or the same lemonade at another store is a great way to decide whether you think it's worth it to buy the lemonade now. Like most heuristics, your brain uses anchoring/adjustment to help you out. It'll usually only cause a problem if someone is actively trying to trick you, like in the example below.



Hey Liz, I'm selling my guitar for \$100.



That's crazy! I'll pay you half of that. \$50.



Ha! It was only worth \$20.



Aw, nuts!



Hey Sunny, I'm selling my guitar for \$1000.



Ha! It was only worth \$20.

That's crazy! I'll pay you half of that. \$500.



Oh no!

Availability

Our next heuristic links back to our discussion of concept representation and learning by association! Specifically, the idea that thinking about concepts more makes their bonds to other ideas stronger in your mind. Recall that this can happen even if the bonds are “incorrect,” like associating dolphins with fish when they are not fish, or tomatoes with vegetables when they are berries. The **availability heuristic** describes how if something comes to mind easily, we assume it is true, important, or common. Ideas that are especially interesting or relevant to us are “loud” in our minds, and we tend to overestimate their relevance to the world. On the other hand, we underestimate the prevalence of boring things we usually don’t think about. Here’s an exercise you can try: list what you believe the top three Google searches were in the past month!

Of course, I don’t know what month you’re reading this, but the odds are your mind went to big news items of the past month or year. However, the top Google searches every month are almost always things like “YouTube,” “Gmail,” “Amazon,” and so on—people just trying to get to very popular and common websites. This is such a mundane process that it usually doesn’t come to mind when we think about what people search on the internet!

Another common example of how the availability heuristic works is when you ask people to think about causes of death. Death is a very dramatic event in most of our minds, and when we think about the ways in which lives can end, we often overestimate the more “dramatic” ways and underestimate the more “boring” ways. You can see an example of this in Figure 8.6, which shows how subjects over and underestimated the frequencies of various causes of death in Sweden².

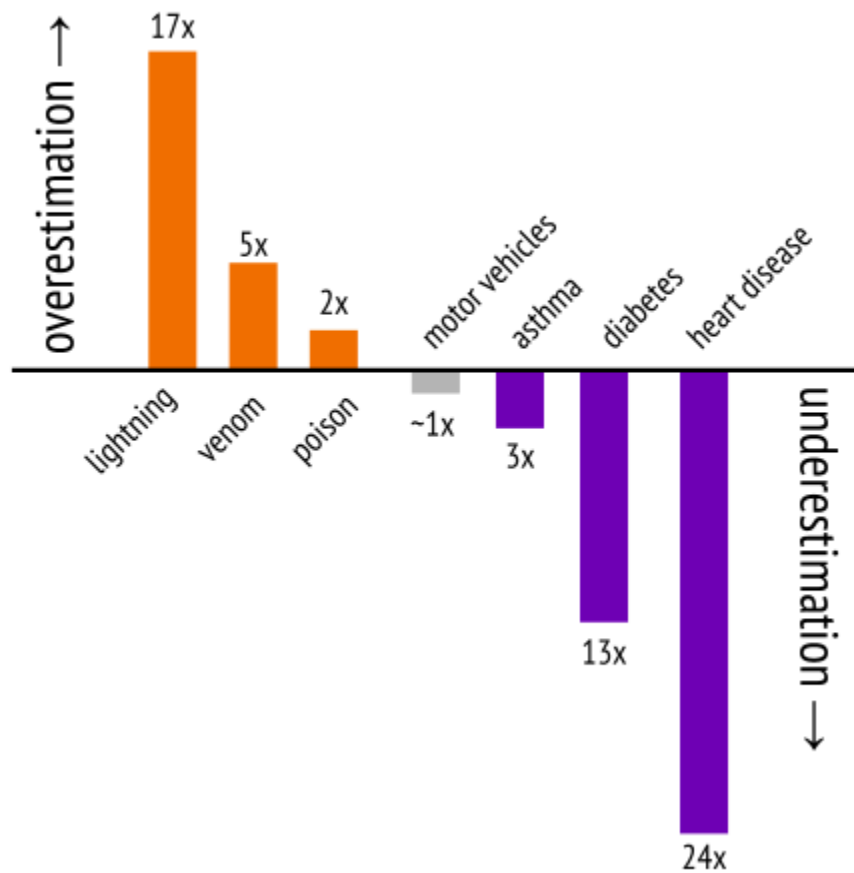


Figure 8.6 Graph of over and underestimated causes of death in Sweden

In the experiment, subjects predicted the frequencies of many different causes of death, and the researchers reported these predicted frequencies alongside the actual frequencies. In Figure 8.6, I've pulled out a few of these causes of death and plotted either how many times larger the subjects' estimation was than the actual value (yellow bars, positive values) or how many times larger the actual frequency was than the subjects' guesses (purple bars, negative values). People tend to believe that dangers like lightning strikes and venomous bites are much more frequent than they actually are, probably because they are very interesting and stick in our brains. Conversely, people usually underestimate the frequency of death from everyday causes like heart disease and diabetes.

Once again, heuristics usually work. The availability heuristic works for you most of the time because most of the time two things are very “loud” in your brain, it's because they *are* frequent, important, or true. Can you guess what the most common pets are in the United States? If you thought of dogs and cats, you'd be right, and it's probably because that's what your brain is used to. In this case, your experience is a good representation of reality.

The Gambler's Fallacy

Agatha is playing a board game with six-sided die. She keeps rolling low numbers...1, 3, 2, 3, 3, 1...a high number is *bound* to come up sometime, right? Isn't it *due*?

We use the **gambler's fallacy** when we assume random events are connected. If we're flipping a coin and it comes up heads six times in a row, it really feels like that means tails *should* be up next. This is a fallacy because it's mathematically wrong. The math of probability dictates that every coin flip has exactly a 50/50 chance of coming up heads or tails, regardless of what happened before. Want to think about something confusing? The following two outcomes of ten coin flips in a row have exactly the same odds of happening (H = *heads*, T = *tails*):

- T H H T H T T T H H
- H H H H H H H H H H

Doesn't that feel wrong? But it's the truth! If you flip a coin ten times, the odds of any specific pattern is $\frac{1}{2} * \frac{1}{2} * \frac{1}{2} * \frac{1}{2} * \frac{1}{2} * \frac{1}{2} * \frac{1}{2} * \frac{1}{2} * \frac{1}{2} * \frac{1}{2}$. Now, that doesn't mean that getting ten heads in a row has the same odds as getting *any* tails in that set of ten coin flips; it just means it has the same odds as any other *exact pattern* of coin flips. If this is easy for you to wrap your head around, great! If not, don't worry! You're a human, and human brains don't work this way. We aren't computers, and we're not good at naturally doing probability math. We use our gut. We use shortcuts.

Humans likely assign meaning and patterns to random events because most natural things in our lives are *not* random. Randomness is an infinite resource; every time you flip a coin or roll a die, all values are available. Rolling a 3 doesn't reduce the number of 3's you could roll. In real life, if you eat five apples from an apple tree, you're now less likely to find another apple on that tree. You've depleted the resource! So, while it's called the gambler's fallacy, it's not really fair to call it a *fallacy*. It's very reasonable for human brains to assume that events follow natural rules. However, if you find yourself playing a game, gambling, or doing anything that involves randomness, it's good to keep in mind not to assume that independent events are linked!

The Hot Hand Fallacy

The **hot hand fallacy** is the gambler's fallacy's mirror image. Sometimes, instead of assuming that random values are a limited resource that need to "even out," we assume that a streak will keep going. The term *hot hand* comes from basketball: when a player is scoring a lot of points, we often believe they are "hot" and will be more likely to keep scoring points. Similarly, when we see streaks in random events, we like to give them meaning. Imagine a slot machine that has given you several small payouts in a row. You might think,

wow, this machine is on a roll! I better stay here! When in truth, streaks are a normal part of random patterns.

Consider this randomly generated list of zeros and ones:

```
0 0 0 0 1 0 0 1 0 1
1 1 1 0 1 0 1 0 0 1
1 0 1 1 0 0 1 0 1 0
0 0 1 0 0 1 1 0 0 0
1 1 0 1 1 1 0 0 0 0
0 0 1 1 0 0 0 1 1 0
0 0 0 0 1 0 1 1 0 1
0 0 0 0 0 1 1 1 0 1
1 1 0 0 1 1 0 0 1 0
0 1 1 0 0 1 1 1 0 1
```

If you look it over, you'll see parts that look "nicely random," where it alternates frequently between 1 and 0, and you'll also see parts where there are longer streaks of zeros or ones. None of it has any meaning; each digit is generated independently. But when patterns like this happen around us, we are very quick to attribute meaning.

Again, this "fallacy" isn't that ridiculous. Many things in real life *do* work in streaks. If you see seven blades of grass, you are very likely to see another. If you dig twelve wells in an area and don't find water in any of them, the area probably doesn't have water. Things in nature are not purely random; they are clumpy, streaky, or follow patterns that have some kind of meaning. When we apply this reasoning to things that are more random (e.g. games, math, or gambling), we can make mistakes – but this does not mean we are being irrational.

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9.

REASONING & PROBLEM SOLVING

Introduction

Humans don't only use shortcuts to come up with quick estimates, as we discussed in the previous chapter. We also try to get away with them for more complex problems! We'll explore humans' difficulties with two specific problems in this chapter: the Wason Card Task and the Monty Hall Problem.

Logic and the Wason Card Task

Do you like logic problems?

Let me clarify. Do you like problems that go, “*if x is true and y is not true, is z sometimes true?*” Some people enjoy these considerably, while other people loathe them with every fiber of their being. What does everyone have in common, though? None of our brains are wired to compute these automatically and accurately (computers, on the other hand, are wired exactly right to compute these automatically). You can teach yourself to consciously solve these problems, but knowing the answer to a logic problem is not like knowing what a chair is. Let's illustrate this with a problem following the pattern I outlined above—an *if-then* question known as the Wason Card Selection Task¹.

Suppose you have a stack of cards. Each card has a letter on one side and a number on the back. There are four cards laid out before you, like in **Figure 9.1**. The cards are supposed to follow a rule: *if* a card has an even number, *then* it must have a vowel behind it.



Figure 9.1 The set-up of the Wason Card Task.

Your task is to examine these cards and make sure that all of them are following the rule – efficiently! Theoretically, of course, you could just flip over every card and see what’s on the other side. But let’s assume you have many, many more of these to check, and if you can figure out how to only check the ones you absolutely have to check, that would be best. So, consider the rule:

If a card has an even number on it,

Then it must have a vowel on the other side.

And determine: which card(s) do you absolutely have to flip over in order to make sure all cards are following the rule?

Give it a try! You may have a few ideas. A common thought process is similar to Liz’s below, in the upper panel of **Figure 9.2**.

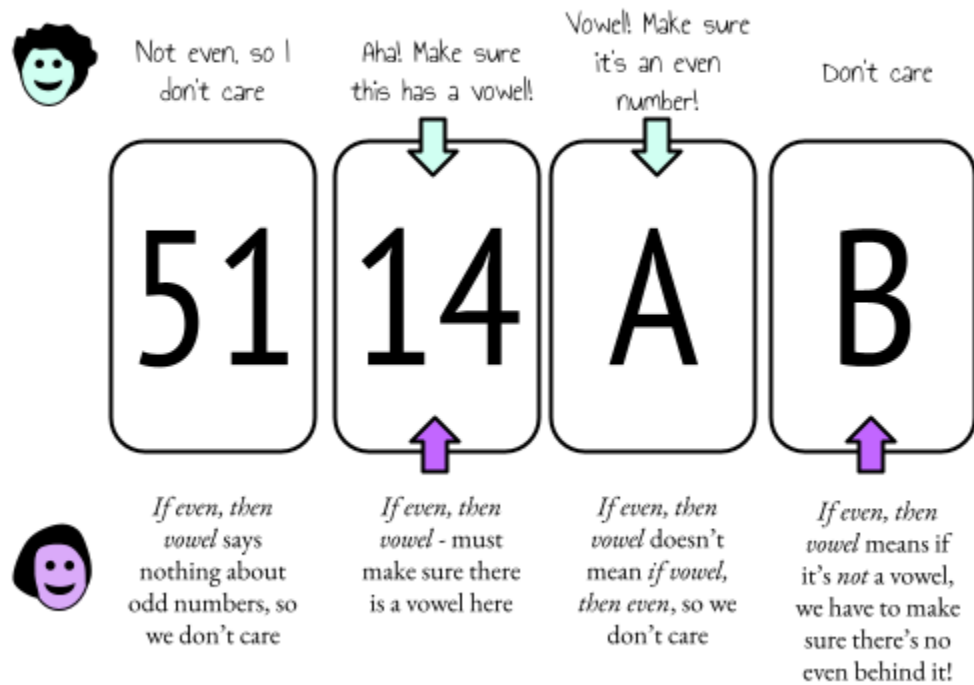


Figure 9.2 Liz's answer (upper panel) to the problem is incorrect, while Agatha's (lower panel) is correct.

If you were thinking similar to Liz, or you selected either the *14* card or the *A* card to check, you would be like most people! In this case, it's not the correct answer – but remember, wrong answers in these sorts of problems don't mean you don't have a well-functioning brain. Humans are not naturally good at this kind of computation. The correct solution is illustrated in the lower panel of **Figure 9.2**.

Did you catch that you need to check the *B* in addition to the *14*? Did you also catch that you don't need to check the *A* at all? Let's briefly go through why Agatha's solution is the correct one. In a logic problem set up like *if X, then Y*, you should definitely check all instances of *X*, to make sure that *Y* goes with them. In this case, you'd be checking even numbers, to make sure there are vowels behind them. But what many people forget about is that every *if X, then Y* statement comes with a secret, implied *if NOT Y, then NOT X* statement. Confused? It might help to substitute in some words. Consider this:

If X, then Y

If Tolly is a rat, then he is an animal.

If NOT Y, then NOT X

If he is NOT an animal, then Tolly is NOT a rat

Both statements about Tolly are true. In fact, in these kinds of logical statements, if the *if X, then Y* statement is true, then the *if NOT Y, then NOT X* statement is also always true! They go together. So, whenever we evaluate an *if X, then Y* claim, we always have to remember to check the *if NOT Y, then NOT X* claim.

Before we leave logic (to go on to everyone's other favorite topic, probability) I would like to pose one more

logic problem. Suppose you are a bouncer at a bar where the law is that no one under the age of 21 may drink alcohol. Your job is to make sure that no one is breaking the law! What you see is detailed in **Figure 9.4**: an adult, a child, someone who ordered an apple juice, and someone who ordered a beer. Who do you absolutely need to check to make sure they are not breaking the law?

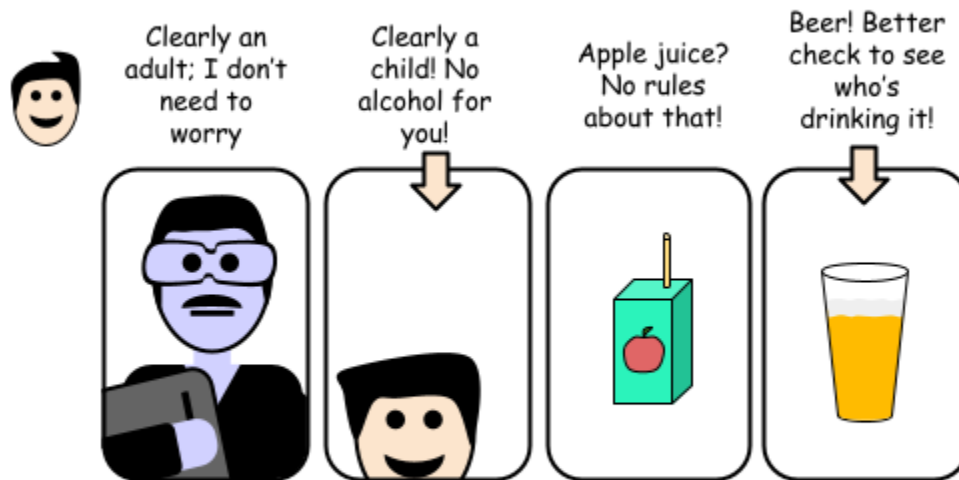


Figure 9.4 Sunny checks the child and the beer-drinker. This is the correct solution.

If you and Sunny had the same thought process, this time, you'd be absolutely right! In fact, more people get this kind of question right². You might even find it downright easy. But, as observant readers may have noted, *this is the exact same problem as the Wason Card Task*. Both questions boil down to “The rule is *if X, then Y*. Do you need to check X? (Yes) Do you need to check Y? (no) Do you need to check NOT-X? (no). Do you need to check NOT-Y? (yes).” In fact, you can even more directly substitute the problems for each other: in the Wason Task presented earlier, the numbers are 51 and 14. You can think of these as ages. The letters are A and B. You can think of these as *apple juice* and *beer*.

Why is the “alcohol rule check” problem so much easier for people than the original Wason Card Task? There are many proposed explanations, but a broad one we'll go with here is simply *context*. Context is the magical glue that makes meaning of things; it allows our brains to stop saying “how do I solve this logic problem” and translating it to something easy and familiar. We're not used to consciously stating *if X then Y*, but we *are* used to thinking about what we're allowed to do and when we're allowed to do it. The context of the bar allows us to skip the part where we try to consciously work out something our neurons have already figured out how to do.

Probability and The Monty Hall Problem

One of the most famous problems in decision science is called the “Monty Hall Problem.” It gets its name from a game show host (Monty Hall) and the show “Let’s Make a Deal,” which started in the 1960s. Here’s the scenario:

You’re looking at three closed doors. You know that behind one of the doors is a brand-new car! But behind the other two doors is...a goat. Assume that you don’t want a goat.

Monty Hall asks you to select a door. Suppose you select Door #1.

“Okay,” says Monty Hall, “You’ve selected Door #1. But before I show you what’s behind Door #1, I want to show you one of the doors you *didn’t* pick!” Suppose Monty Hall then opens Door #3, and reveals a goat behind it.

“Now,” he says, “You have a choice. Do you want to *stay* with Door #1, or do you want to *switch* to Door #2?”

You can examine this scenario as illustrated in Figure 9.6.

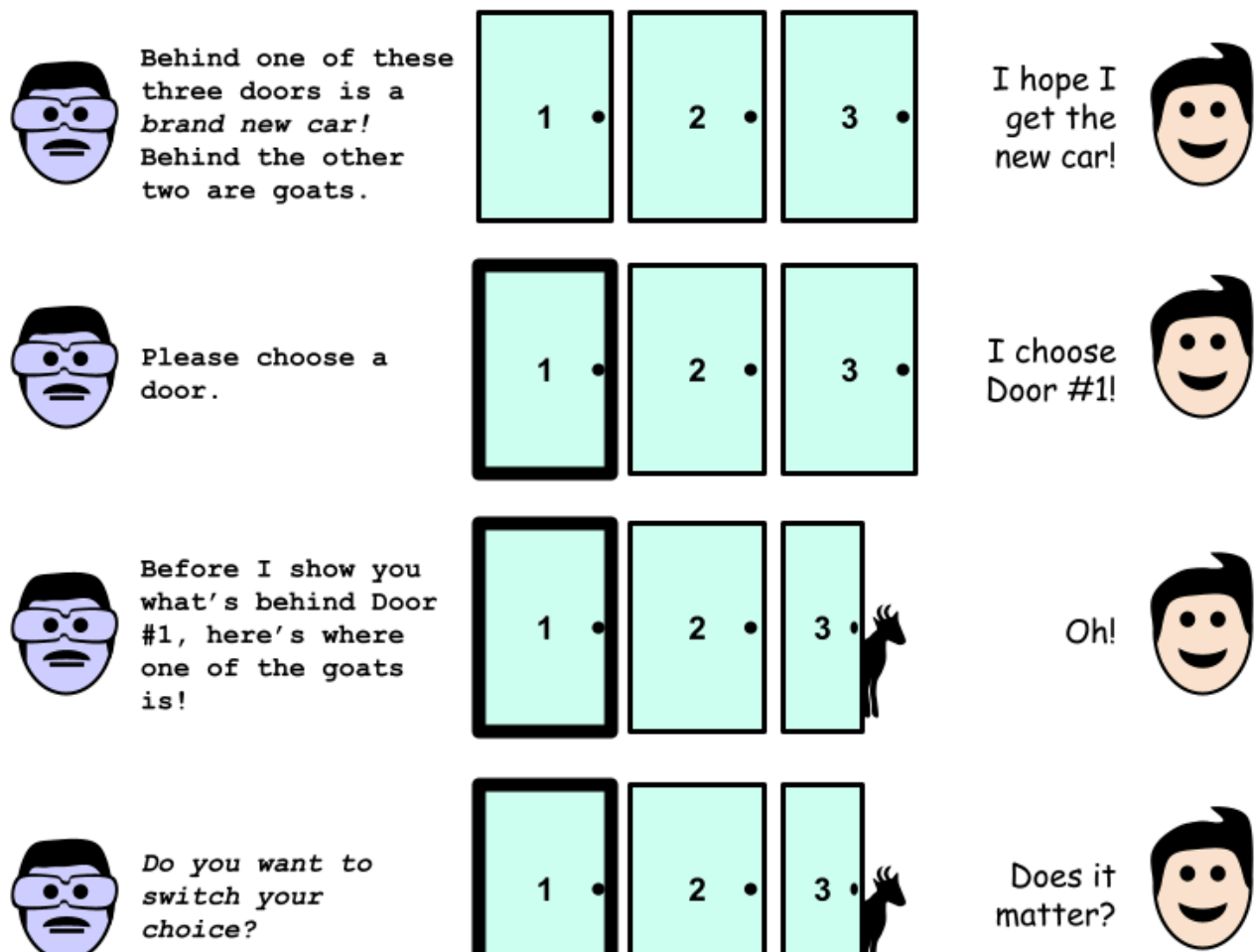


Figure 9.5 Dr. Gotcha confronts Sunny with the Monty Hall Problem.

Before we resolve the scenario, we should stop to consider the question of switching. Should you switch from Door #1 to Door #2? Should you stay with Door #1? Does it even matter?

Many people have a gut feeling that it *should not matter* whether you switch or stay. When you make your initial choice, you have a $\frac{1}{3}$ chance of landing on the car. Opening one of the doors doesn't go back in time and change the probability of that initial choice. So, in our minds, the probability of either Door #1 or Door #2 having the car is *still* $\frac{1}{3}$:

But this is wrong! In fact, you are more likely to get the car if you switch. In fact, you are *twice* as likely! The “correct” answer to the Monty Hall Problem is to always switch your initial choice.

Does that feel right? If it doesn't, don't worry. It is true that when you make your initial selection, the odds of you getting the car are one in three: $\frac{1}{3}$. Learning the contents of one of the doors you didn't pick doesn't change that. However, it *does* change the probability of the *other* door! When you make your initial choice, everything is random. But once Monty Hall chooses to show you a goat, that is not a random choice. Let's look at it this way, with every possibility laid out:

- If Door #1 contains the car, Monty will show you one of goats behind one of the other doors.
- If you STAY, you will get the car!
- If you SWITCH, you will get a goat.
- If Door #2 contains the car, Monty will show you the goat behind Door #3
- If you STAY, you will get a goat.
- If you SWITCH, you will get the car!
- If Door #3 contains the car, Monty will show you the goat behind Door #2
- If you STAY, you will get a goat.
- If you SWITCH, you will get the car!

One out of three times, you will have chosen the car at random. Two out of three times, you will have chosen one of the goats.

In other words:

- You'd only get the car by switching if you picked a goat initially
- You initially pick a goat $\frac{2}{3}$ of the time!

If this has sunk in for you, fantastic! If you are like me though, you might see all the scenarios laid out, see the math, and say “yes, I understand why this is true,” but maybe not be able to really *feel* it in your gut. If you feel this way, you are not alone. This is why the Monty Hall Problem is famous: the solution just doesn't *feel* right.

Chapter 9 Wrap-Up

Both problems we discussed are examples of how humans naturally struggle when directly faced with logic or probability. This does not mean we are stupid; nor does it mean that you are abnormal if you get good at these things. Rather, all it means is that we are not computers. Computers are quite literally built on logic. Our organic, squishy brains, on the other hand, are built on organic, squishy chaos. We learn better through experience and association. If you did the Monty Hall Problem a hundred times, it would probably start to sink in that switching is better than staying. (Can we link to a site that does that? Or is that not OER?)

References

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10.

VALUE

Introduction

Do you prefer chocolate or vanilla? Scary movies or romantic movies? Cats or dogs?

You may have simple, clear answers for each of those above questions. But, you may not. You may be thinking, “Why do I have to pick just one? They aren’t mutually exclusive! They’re not even opposites!” You may even be thinking, “I hate all of those things!” The representation of all this information in your brain is what we call **value**.

Types of Value and the Brain

Consider Sunny, who’s at a bakery, trying to decide between a chocolate donut and a strawberry donut. **Figure 10.1** lays out the differences between them both in terms of their monetary value and in terms of Sunny’s own personal value:



Figure 10.1 Sunny’s subjective value for the strawberry donut is higher, even though the chocolate donut is cheaper.

Based on monetary value alone, the chocolate donut is the better option—Sunny gets the same amount of donut, but for less money. However, when you take into account Sunny’s own personal preferences which favor strawberry, he might go for the strawberry instead. Which donut Sunny chooses will be the result of

a computational process that takes into account both the monetary value and the **subjective value** (how much Sunny in particular cares about the donut).

There is a broad region of your brain that cares about these sorts of things, and it's one we've already discussed in this book...the prefrontal cortex! To be sure that this chunk of brain tracks value, we can measure the activity of prefrontal neurons and see if they correlate with the value of things you're seeing. Some neurons in Sunny's prefrontal cortex might look like **Figure 10.2**.

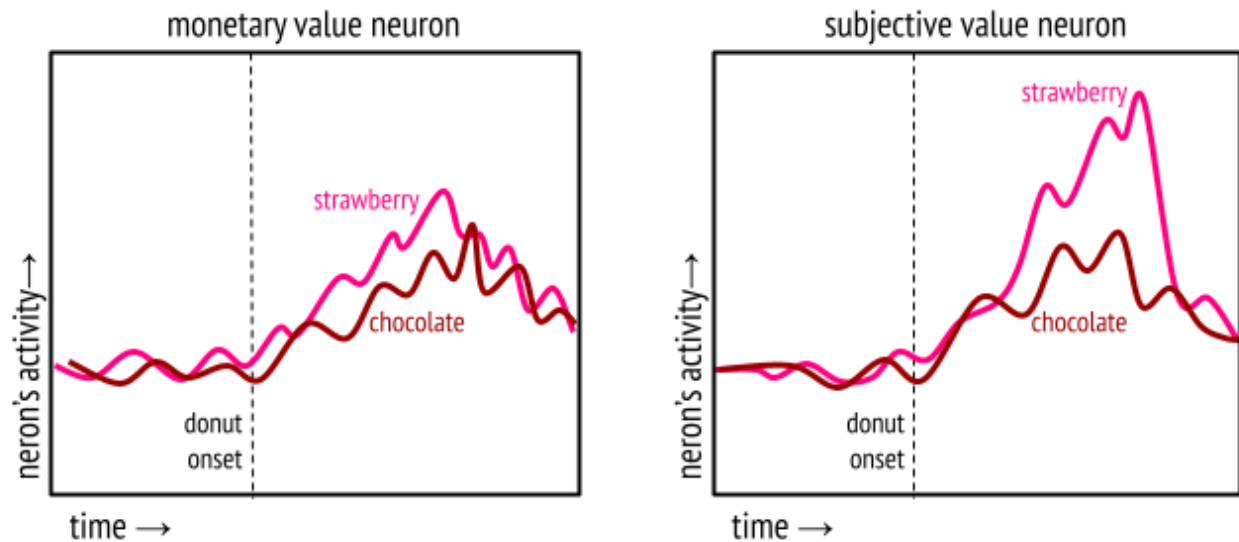


Figure 10.2 Neurons tuned to monetary value (left) and subjective value (right).

The first panel shows a “monetary value” neuron. This neuron gets excited upon seeing the chocolate donut, because it has some value. It gets a little more excited for the strawberry donut, because its value is a little higher. The second panel shows a “subjective value” neuron. Upon seeing the chocolate donut, this neuron gets a little excited. But when Sunny sees the strawberry donut, the neuron gets a lot more excited. These patterns of behavior are typical for neurons in places like the orbitofrontal cortex (citation) and the ventromedial prefrontal cortex (citation).

So, our brains care about not only the objective value of things in our world, but also subjective value. The latter is a lot more common than the former; you probably care about a lot more things than you know the prices of, off the top of your head. It looks like the idea of value in the brain may not be so straightforward...

Let's look at some more factors that affect your perception of value!

Expected Value

Would you rather have \$10, or spin a wheel for a 10% chance at winning \$100? You might be the cautious,

steady kind of person. You might take the \$10 because it's a sure thing, and it's \$10 you didn't have before. Or, you might enjoy a bit of a risk! The thrill of spinning the wheel and the potential of getting a big payout might be worth not getting that \$10. Whichever option you prefer, I'm willing to bet my own \$10 that you don't think these two choices are the same.

According to **expected value**, they are!

Expected value is the average value of an option over infinite time. In the case of the problem above, the expected value of the first option is \$10, because it's just a straightforward offer of \$10. If you take that option 10 times, you'll end up with \$100. But the expected value of the second option is *also* \$10, because if the odds of getting \$100 are 1 in 10, and you spin the wheel 10 times, you'll also probably end up with \$100. In other words, if you multiply the probability (10%) by the total (\$100) you get the same number as the first option (\$10).

Mathematically, a certain \$10 is the same as a 10% chance of \$100, if you run through this problem infinitely many times. But do they *feel* the same to you? How can we quantify exactly how different these two options feel to us, even if we can consciously see that the math works out to give us the same reward over time?

Consider the following list of questions. Would you rather...

- Gain \$50 or take a 10% chance at gaining \$100?
- Gain \$50 or take a 25% chance at gaining \$100?
- Gain \$50 or take a 50% chance at gaining \$100?
- Gain \$50 or take a 75% chance at gaining \$100?
- Gain \$50 or take a 90% chance at gaining \$100?

Your answer likely changed as the odds of getting the \$100 increased. We can calculate the expected values of each choice as:

- \$50 vs \$10
- \$50 vs \$25
- \$50 vs \$50
- \$50 vs \$75
- \$50 vs \$100

If you always make choices perfectly in line with expected value, this is what a graph of your choices would look like **Figure 10.3**:

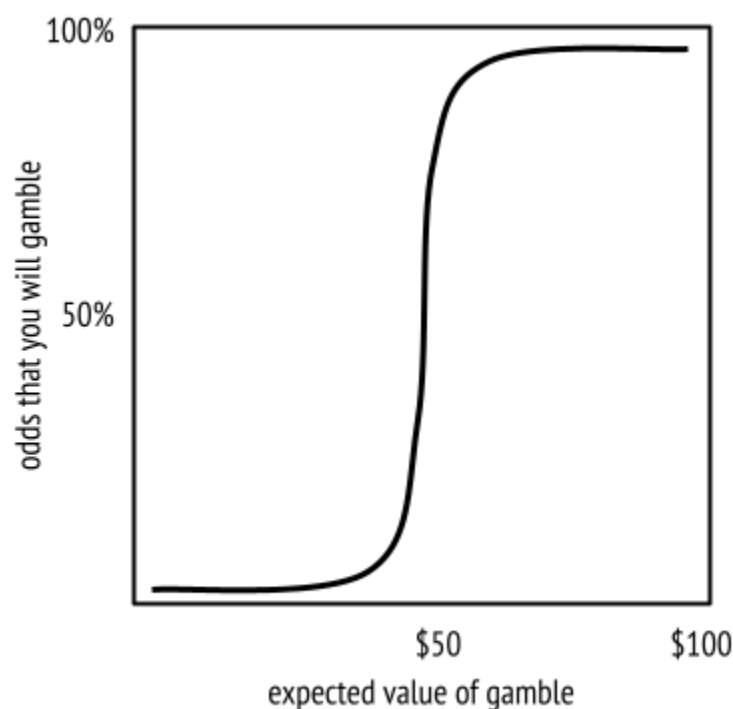


Figure 10.3 Odds of you gambling if you only choose to gamble when the expected value of gambling is higher than \$50

According to mathematically rational behavior, you should always take the \$50 when the expected value of the gamble is lower than \$50, you should always take the gamble when the expected value of the gamble is higher than \$50, and it's random chance which one you should take when both expected values are equal. Is this how you chose? Perhaps! But life is rarely populated with such clean choices to make. Indeed, humans tend to pay some mind to expected values, but on average we don't act "perfectly" mathematically. We all behave a little differently.

Delays, Distractors, and Diminishing Returns

The same amount of something can have a different value in different contexts. An ice-cold glass of water might have more value to you after a run in the hot sun than after a movie in an air-conditioned theater during which you drank a soda. An extra quarter might be very valuable to you in a cash-only establishment, but not much to you otherwise. A \$30 shirt may not seem like a very good deal, unless it is next to a similar, \$40 shirt. This last example illustrates the **decoy effect**: when you are more likely to select an option when there is a worse option there for you to compare it to. In all these cases, context is key.

What about your initial state of being as a form of context? When you have \$10, getting another \$10 is great! When you have \$1000, getting \$10 is still nice, but not quite as impactful as it was when you only had

\$10. This specific phenomenon is an example of **diminishing returns**: the more you get of something, the less valuable each addition of it is.

Having to wait for a reward is another factor that might diminish its value. Consider the following choices. Would you rather get...

- \$50 now or \$100 tomorrow?
- \$50 now or \$100 next week?
- \$50 now or \$100 next month?
- \$50 now or \$100 next year?
- \$50 now or \$100 in ten years?

Mathematically, \$100 is always greater than \$50. So there should never be a reason not to choose the \$100, right? Not unless the *delay* has some effect on value. Indeed, many people find that it does.

Winning and Losing

All our examples so far have to do with the potential to gain money. What about losing money? Do the same rules govern our cognition surrounding that?

Consider the following questions. Would you rather...

- Lose \$50 or take a 10% chance at losing \$100?
- Lose \$50 or take a 25% chance at losing \$100?
- Lose \$50 or take a 50% chance at losing \$100?
- Lose \$50 or take a 75% chance at losing \$100?
- Lose \$50 or take a 90% chance at losing \$100?

You might find that your answers to this set of questions follow a different pattern than your answers to the “gain” version of the same questions. Often, people find the idea of just accepting a loss of \$50 to be extremely disagreeable. Indeed, people tend to be riskier with potential losses than potential gains (reference), because “losses loom larger than gains.” Figure 10.4 shows the relationship between gains and subjective value, compared to the relationship between losses and subjective value. The amount that people tend to be pleased about gaining something is often lower than the amount that people tend to be upset about losing that same thing (reference).

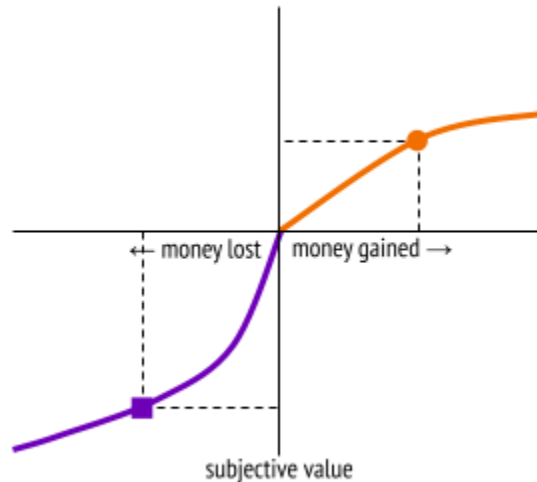


Figure 10.4 An amount of money gained causes a smaller change in subjective value than the same amount lost

We don't just hate to lose things. We also hate for things to “go to waste.”

Suppose Sunny is at a restaurant he's been to before. He usually gets spaghetti and meatballs, but today's special is spinach alfredo. He's curious about it and orders it, but when it arrives, he tries it and realizes he hates it. Sunny paid \$17 for the spinach alfredo. Should Sunny just eat it, or should he order the spaghetti and meatballs for an additional \$16?

Many people in this and similar situations will find themselves suffering through the thing we don't like, *because we already paid for it*. Even if we didn't eat the spinach alfredo, we wouldn't go through the trouble of ordering a whole new dish. If you can relate to that feeling, you've probably employed the **sunk cost fallacy** before: assigning value to costs we have already paid

Logically, Sunny is never getting that \$17 back. It's gone. His decision to pay \$16 more for a meal he will actually eat will not change that. But Sunny feels stress from the money already spent—the *sunk* cost. Sunny is also the kind of person to feel more pressured to attend an event if he's pre-purchased tickets, to finish a book that he doesn't like because he's already devoted time to it, or to go through with a project even if it's started poorly. In short, he doesn't like to cut his losses.

GLOSSARY

abstract representation

algorithm

cognitive map

cortex

dualism

limbic system

monism

pictorial representation